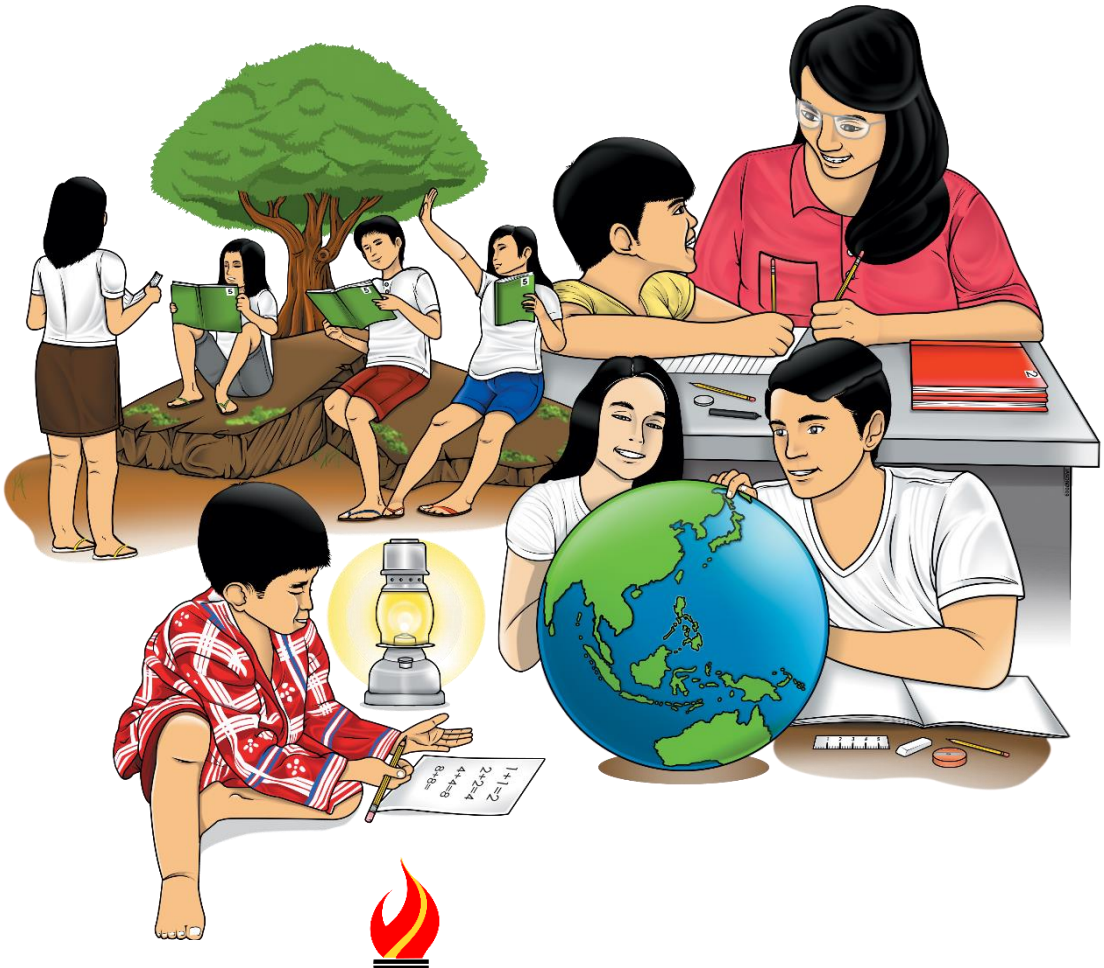
Senior High School

DepED
DEPARTMENT OF EDUCATION

General Physics 1

Quarter 1 – Module 1

Measurements, Vectors, Relative Motion, Newton's Laws of Motion, Work, Forces, Center of Mass and Collisions



ALTERNATIVE DELIVERY MODE
ADM

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General Physics 1 – Grade 12
Alternative Delivery Mode
Quarter 1 – Module 1: Measurements, Vectors, Relative Motion, Newton's Laws of Motion, Work, Forces, Center of Mass and Collisions
First Edition, 2020

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Introductory Message

For the facilitator:

Welcome to the General Physics 1 Alternative Delivery Mode (ADM), this module was collaboratively designed, developed and reviewed by educators both from public and private institutions to assist you, the teacher or facilitator in helping the learners meet the standards set by the K to 12 Curriculum while overcoming their personal, social, and economic constraints in schooling.

This learning resource hopes to engage the learners into guided and independent learning activities at their own pace and time. Furthermore, this also aims to help learners acquire the needed 21st century skills while taking into consideration their needs and circumstances.

As a facilitator you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Furthermore, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

The hand is one of the most symbolized part of the human body. It is often used to depict skill, action and purpose. Through our hands we may learn, create and accomplish. Hence, the hand in this learning resource signifies that you as a learner is capable and empowered to successfully achieve the relevant competencies and skills at your own pace and time. Your academic success lies in your own hands!

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning resource while being an active learner.

This module has the following parts and corresponding icons:

What I Need to Know

This will give you an idea of the skills or competencies you are expected to learn in the module.

What I Know

This part includes an activity that aims to check what you already know about the lesson to take. If you get all the answers correct (100%), you may decide to skip this module.

What's In

This is a brief drill or review to help you link the current lesson with the previous one.

What's New

In this portion, the new lesson will be introduced to you in various ways such as a story, a song, a poem, a problem opener, an activity or a situation.

What is It	This section provides a brief discussion of the lesson. This aims to help you discover and understand new concepts and skills.
What's More	This comprises activities for independent practice to solidify your understanding and skills of the topic. You may check the answers to the exercises using the Answer Key at the end of the module.
What I Have Learned	This includes questions or blank sentence/paragraph to be filled in to process what you learned from the lesson.
What I Can Do	This section provides an activity which will help you transfer your new knowledge or skill into real life situations or concerns.
Assessment	This is a task which aims to evaluate your level of mastery in achieving the learning competency.
Additional Activities	In this portion, another activity will be given to you to enrich your knowledge or skill of the lesson learned. This also tends retention of learned concepts.

Measurements
Quarter 1 – Week 1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topics about measurements. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module has three lessons, namely:

- Lesson 1 – Solving Measurement Problems
- Lesson 2 – Uncertainties in Measurement
- Lesson 3 – Vectors and Vector Addition

After going through this module, you are expected to:

1. solve measurement problems involving conversion of units and expression in scientific notation;
2. estimate the uncertainty of a derived quantity from the estimated values and uncertainties of directly measured quantities;
3. differentiate vector and scalar quantities;
4. classify the physical quantities as scalar and vector quantity; and
5. perform addition of vectors.

What I Know

Directions: Complete the table by converting the given unit of km to m. Note: 1 km = 1000 m

Metric Unit	Measurements		
Kilometer (km)	93.5	53.78	16
Meter (m)			

Lesson 1 Solving Measurement Problems

When measurement is made, we compare a known value which is commonly called ***standard or acceptable measurement*** against an unknown measurement. To understand measurement, there must be a number to quantify measurement and a unit which serves as reference of the measurement.

In doing measurement, the foremost objective is to measure accurately. But uncertainty in measurement is always there. Uncertainty in measurement does not mean that the measurement made is wrong.

What's In

How many units do we have for 1 foot or 12 inches ruler?
How many millimeters (mm), centimeters (cm), inches (in), foot (ft)?

Classify the given quantities as Metric System or British System of measurement: 3 yards of cloth, 5 kg of apple, 6 hours watching TV, 10 inches shoes, 15 miles walking and 60 km/h speed of Typhoon Yolanda. How do we convert metric measurements? Remember this passage.

King	Henry	Doesn't	Usually	Drink	Chocolate	Much
Kilometer	Hectometer	Decameter	Unit	Decimeter	Centimeter	Millimeter

What is It

Scientific notation

For convenience, a large or small measurement in Physics use the scientific notation, which is sometimes called the power of ten or exponential notation.

$$M\ x\ 10^n$$

where: M - the number (1 digit left of decimal)

n - the exponent; positive when the decimal point is moved to the left and negative when decimal point is moved to the right).

Rules to determine number of significant figures

1. All **non-zero** digits are significant.
   ex. 12.33 - **4** significant figures
   0.267 -**3** significant figures
2. Zeroes in a measured quantity are significant figures.
   ex. 5.0020 -**5** significant figures
   1.30 -**3** significant figures
3. Zeroes to locate the decimal point are not significant figures.
   ex. 0.0045 -**2** significant figures
   0.000002 -**1** significant figures
4. Numbers with trailing zeros to the right is not significant, if the number is whole number.
   ex. 200 -**1** significant figures
   7800 -**2** significant figures

Physical Quantity is the measurable property of an object like its length, mass and time. *Derived Quantity* is the combination of two or more fundamental quantity.

There are two systems of measurement which are **British or English system** and **Metric system**. The English units of measure which is sometimes called the FPS system uses foot for the length, pound for the mass and second for the time. The metric units of measure, also called SI system or the international system of measurement, consists of standard units and prefixes that are multiple of ten. It includes the MKS system (meter – kilogram – seconds) and the CGS system (centimeter – gram – seconds) for the length, mass, and time.

Conversion of Units

1. Conversion of unit using one conversion factor.
   Example: Convert 5 feet to inches

$$5\ feet\ x\ \frac{12\ inches}{1\ foot} = 60\ inches$$

2. Conversion of unit using more than one conversion factor.
   Example: Convert 7 meters to yard

$$7\ meters\ x\ \frac{3.28\ feet}{1\ meter} x\ \frac{1\ yard}{3\ feet} = 7.65\ yards$$

3. Conversion of unit which contains an exponent. **Note:** Temporarily ignore the exponent in the given unit and select the needed conversion factor. Then get the squared value of the needed conversion factor.
   Example: Convert 4 ft2 to inches2
   conversion factor:
   1 foot = 12 inches ; $(1\text{foot})^2 = (12\text{ inches})^2$
   required conversion factor: $1\ ft^2 = 144\ inches^2$

$$4\ ft^2\ \text{x}\ \frac{144\ inches^2}{1\ ft^2} = 576\ inches^2$$

4. Conversion of unit in both numerator and denominator. **Note:** Change the unit in the numerator or denominator one at a time.

Example: Convert 15 km/hr to m/s

$$15\ \frac{km}{hr} \text{ x } \frac{1000\ m}{1\ km} \text{ x } \frac{1\ hr}{3600\ s} = 4.17\ m/s$$

5. Conversion of unit using the prefixes for the power of ten.

Example: Convert 5 hm to dm

$$5\ hm\ x\ \frac{10^2\ m}{hm}\ x\ \frac{dm}{10^{-1}\ m} = 5000\ dm\ or\ 5\ x\ 10^3\ dm$$

What's More

Directions: Convert the following given:

1. 25 Liter to cm^3
2. 18, 000 cm^2 to m^2

What I Have Learned

How do you convert one unit factor to another unit factor?

_____

What I Can Do

Directions: Solve the following problems.

1. The speed of a car is 60 Km/hr. what is the speed of the car in m/s?
2. A healthy person must drink at least 2 liters of water a day. How many gallons of water must a healthy person drink in a day?

Assessment

Scientific Notation.

A. Write the given value in scientific notation using 3 significant figures.

1. 15, 000, 000 = _____
2. 0.0002027 = _____
3. 123.45 = _____
4. 0.0000004 = _____
5. 12,507 = _____

B. Write the given value in Standard form.

1. 2.37×10^4 = _____
2. 3.458×10^2 = _____
3. 2351×10^{-3} = _____
4. 2.26×10^2 = _____
5. 3×10^{-3} = _____

Conversion Factor. Convert the following. Use separate sheet of paper for your solution and answer.

1. 30 miles/hr to ft/sec.
2. 230 m to yards
3. 36, 000 seconds to minutes

Lesson 2 Uncertainties in Measurement

All experimental uncertainty is due to either random errors or systematic errors. Random errors are statistical fluctuations (in either direction) in the measured data due to the precision limitations of the measurement device.

What's In

Directions: Read the statements below and answer what is asked in each number.

1. Five (5) senior high school students measured a 5 peso-coin. The measurements are 2.74 cm, 2.78 cm, 2.78 cm, 2.76 cm, 2.90 cm. Give your interpretation about the result of q findings.
2. Given the following data, solve for the mean and variance of the following.

   8, 9, 10, 10, 10, 11, 11, 11, 12, 13

What is It

Estimating Errors in Measurement

To increase the precision and accuracy of a single measurement, the *Least Count Concept (LCC)* is used to estimate the error. It is called LCC because the measuring instrument itself offers the smallest change or margin of error in its scale.

Statistics in Multiple Measurement is used to increase the precision and accuracy of a multiple measurement, statistics helps in estimating and reducing the errors. Average is used to get the exact value in multiple measurement. The more trial measurement, the more accuracy of the result. Variance is use to get the exactness of multiple measurement.

$$average\ measurement\ (\underline{x}) = \frac{sum\ of\ the\ multiple\ measurement}{total\ number\ of\ measurement}$$

Uncertainty of Derived Quantity

Derived quantity like area, volume or velocity are perform in Physics experiment. All fundamental measurements contain the actual measurements and their uncertainties.

$$\Delta u = \left[\frac{a}{c} + \frac{b}{d}\right] x\ m$$

NOTE: derived measurement & change of uncertainty = m ± Δ

What's More

where: Δu = total change of uncertainty
a & b = change of uncertainty in two measurements
c & d = two actual measurements
m = product or quotient of the actual measurements

Example: Consider: Five (5) students measured the diameter of 25 centavo coins.

No. of trials	Measured Diameter	$(x - \underline{x})$	$(x - \underline{x})^2$
1	2.14 cm	(2.14 – 2.08)	0.0036
2	2.09 cm	(2.09 – 2.08)	0.0001
3	2.00 cm	(2.00 – 2.08)	0.0064
4	2.11 cm	(2.11 – 2.08)	0.0009
5	2.05 cm	(2.05 – 2.08)	0.0009
		∑ →	0.0119

a) Find the mean.

$$\underline{x} = \frac{2.14 + 2.09 + 2.00 + 2.11 + 2.05}{5} = 2.08\ cm$$

b) Solve for the variance.

$$s^2 = \frac{\sum(x - \underline{x})^2}{n - 1} \qquad s^2 = \frac{0.0119}{5 - 1} = 0.003$$

Interpretation: since the variance is small, there is precision in the measurements made by the students.

What I Have Learned

How do we discard uncertainties in measurement?

_____

What I Can Do

Directions: Solve the problem below on a separate sheet of paper.

Six senior high school students recorded the time to cover certain distance. Determine the average and variance and interpret the result.

No.	Measurement (x)
1	28 sec
2	22 sec
3	18 sec
4	23 sec
5	25 sec
6	19 sec

Assessment

Directions: Solve the given the problem. Show your solution in a separate sheet of paper.

A rolling ball covers a distance of 12.6±0.5 centimeters in 2.5±0.02 sec. What is the speed of the ball in that distance?

Lesson 3 Vectors and Vector Addition

We come into contact with many physical quantities in the natural world on a daily basis. For example, things like time, mass, weight, force, and electric charge, are physical quantities with which we are all familiar. We know that time passes and physical objects have mass. Things have weight due to gravity. We exert forces when we open doors, walk along the street and kick balls. We experience electric charge directly through static shocks in cold temperature and through using anything which runs on electricity.

There are many physical quantities in nature, and we can divide them up into two broad groups called **vectors** and **scalars**.

What's In

Directions: Choose the best answer on the given words inside the box. Write the letter before the number on the blank provided.

7 cm, South	B. 5 cm, West	C. 10 cm, East	D. 2 cm, North

_____ 1. 7 cm, East + 3 cm, East
_____ 2. 12 cm, North + 10 cm, South
_____ 3. 15 cm East + 20 cm, West
_____ 4. 3 cm, South + 4 cm, South

What is It

VECTORS AND ITS PROPERTIES

Vectors – quantities expressed by magnitude and direction. ($\vec{\ }$).
Examples. Force, Velocity, Displacement

Scalar – quantities which are completely specified by their magnitude or numerical value.
Examples. length, mass, area, volume time, density

Example: Identify the following whether it is vector or scalar:

A. 50 m/sec, east - **vector**
B. 8 mi, north - **vector**
C. 57 bytes - **scalar**
D. 2000 calories - **scalar**

Parts of a Vector

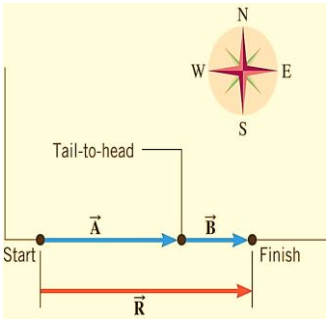
1. Arrowhead – indicates the direction of the vector.
2. Length of the arrow – represents the magnitude of the vector.
3. Tail – refers to the origin of the vector.

RESULTANT DISPLACEMENT VECTOR $\vec{R}$ – a single vector representing the sum of 2 or more vectors.

Methods in finding $\vec{R}$

1. Graphical method
   a. Polygon method (head to tail)
   b. Parallelogram method
2. Analytical method
   a. Pythagorean theorem
   b. Component method

Illustration on how to connect vectors. (Addition or subtraction)



ADDITION OF VECTORS (same direction)
A = 3 cm East B = 2 cm East R = 5 cm East
SUBTRACTION OF VECTORS (opposite direction)
C = 5 cm west D = 6 cm east R = 1 cm east

Given:
A = 6 cm, E B = 2 cm, N E = 2 cm, E
C = 2 cm, W D = 9 cm, W

RULES IN VECTOR ADDITION (FINDING THE RESULTANT VECTOR)

When **two given vectors** are acting on the **same direction.**

- To find the magnitude and direction of the resultant.
- Add the magnitude of the two vectors to get the resultant.
- For the direction, use same direction of the given vectors.

1. A + E = 6 cm, E + 2 cm, E = **8 cm, E**

When **two given vectors** are acting in the **opposite direction**.

- To find the magnitude and direction of the resultant.
- Subtract the magnitude of the two vectors to get the resultant.
- For the direction, follow the direction of the larger magnitude.

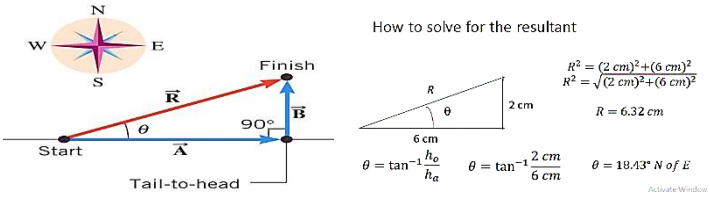
2. A + D = 6 cm, E + 9 cm, W = **3 cm, W**
   Note: notice that the magnitude **3 cm** is still positive, in vector the **negative sign** will be represented by the direction (WEST in the x-axis and SOUTH in the y-axis)

When **two vectors** forming an **angle 90 °** with another.

- To find the magnitude and direction of the resultant.
- The diagonal form by the two vectors is the resultant.
- Use Pythagorean theorem (R^2 = $\mathbf{c^2 = a^2 + b^2}$) to find
- Magnitude of the resultant.
- For the direction: use SOHCAHTOA in TRIGONOMETRY

3. A + B = 6 cm, E + 2 cm, N
   Solution: Observe that vector B lies on x-axis; vector E lies on y-axis.

Note: In solving resultant vector in 2 dimensions, consider also the angle of reference along the x-axis.



When **two vectors** form an **oblique triangle.**

- To find the magnitude and direction of the resultant.
- The diagonal form by the two vectors is the resultant.
- Use cosine and sine law to find the magnitude and direction of the result

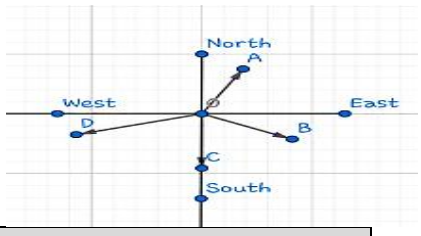
NOTE: Use the component method for this procedure.

What's More

Component Method – this method starts by resolving the vectors to their horizontal and vertical components. The vector sum of the horizontal components represents the horizontal component of the resultant while the vector sum of the vertical components represents the vertical component of the resultant.

Example: Find the vector sum of the four coplanar vectors acting at O as shown in the figure below.

Vectors: A : $150\ N, 62°\ N\ of\ E$
B : $180\ N, 23°\ S\ of\ E$
C : $130\ N, S$
D : $125\ N, 25°\ S\ of\ W$



Vectors	Horizontal Component (x-axis)	Vertical Component (y-axis)
A	$150\ N\ cos\ cos\ 62° = 70.421 N, E$	$150\ N\ sin\ sin\ 62° = 132.442\ N, N$
B	$180\ N\ cos\ cos\ 23° = 165.691, E$	$150\ N\ sin\ sin\ 23° = 70.332\ N, S$
C	$130\ N\ cos\ cos\ 270° = 0$	$130\ N\ sin\ sin\ 270° = 130\ N, S$
D	$125\ N\ cos\ cos\ 25° = 113.288\ N, W$	$125\ N\ sin\ sin\ 25° = 52.827\ N, S$
∑	$\Sigma Fx = 122.824\ N, E$	$\Sigma Fy = 120.717\ N, S$

Magnitude and Direction of R:

$$R^2 = \Sigma Fx^2 + \Sigma Fy^2$$

$$R^2 = (122.824)^2 + (120.717)^2$$

$$R = \sqrt{29\,658.32907}$$

$$R = 172.216\ N$$

For angle of reference (along x-axis)

$$\frac{\Sigma Fy}{\Sigma Fx} = \frac{120.717\ N}{122.824\ N} = 0.983$$

$$\theta = 0.983$$

$$\theta = 44.508°\ S\ of\ E$$

Therefore, the **resultant vector** is $172.216\ N, 44.508°\ S\ of\ E$**.**

What I Have Learned

Differentiate vector from scalar.

What I Can Do

Directions: Solve the problem. Show your solution in a separate sheet of paper.

A tennis ball rolls 10 meters east when it hits a stone that turns the ball 80 ° N of E rolling for another 15 m. Calculate magnitude and direction of the resultant displacement of the tennis ball.

Assessment

Directions: Show your solution in a separate sheet of paper.

A car moves 5 km east and turns 45 ° N of E for another 3.25 km before coming to stop. Find the magnitude and direction of the resultant displacement of the car.

Kinematics: Motion Along a Straight Line
Quarter 1 – Week 2

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Kinematics: Motion Along a Straight Line. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module has three lessons, namely:

- Lesson 1 – Motion
- Lesson 2 – Motion Graphs
- Lesson 3 – Uniformly Accelerated Motion

After going through this module, you are expected to:

1. describe motion;
2. interpret velocity vs. time and acceleration vs. time graphs; and
3. convert a verbal description of a physical situation involving uniform acceleration in one dimension into a mathematical description

What I Know

Directions: Read each statement carefully. On the space provided before each number, write TRUE if the statement is correct. Otherwise, write FALSE.

_____ 1. Accelerating objects must be changing their speed.
_____ 2. Displacement time graph of a body moving with uniform velocity is always a straight line.
_____ 3. The slope of a position-time graph of an object gives the speed of the object
_____ 4. In a position vs time graph, the vertical axis represents the position of the object.
_____ 5. The acceleration due to gravity has a magnitude of 9.8m/s2.
_____ 6. When a ball is thrown straight up with no air resistance, the acceleration at its highest point is zero.
_____ 7. If a train is slowing down, it is accelerating.
_____ 8. A Ferris wheel turning at a constant speed is accelerating.
_____ 9. Displacement is the distance between an object's final and initial positions. It must also include the direction of the change in position.
_____ 10. A skydiver jumping out of a plane is an example of uniformly accelerated motion.

Lesson 1 Motion

After accomplishing your week 1 module in General Physics 1, you now understand the basic mathematical concepts in Physics. You have learned about measurements and the difference between vectors and scalars. The next phase of your lesson will be the understanding of the concept of "motion". In this module, you are going to apply what you have learned in the first module. Different activities and tasks were prepared to guide you in establishing an understanding of what motion is.

What's In

Illustrate the equations that describe the physical quantities of motion: distance, displacement, velocity and acceleration.

What is It

Kinematics is the science of describing the motion of objects using words, diagrams, numbers, graphs, and equations. Kinematics is a branch of mechanics. The goal of any study of kinematics is to develop sophisticated mental models that serve to describe the motion of real-world objects.

What is motion? When can we say that a body is in motion?

Motion is defined as the movement of an object. It is also defined as a continuous change in position with respect to a certain frame of reference. There are three types of motion which are **rectilinear motion** where the motion of an object travels in straight path, **curvilinear motion** where the motion of an object travels in a curved path and **angular motion** where the motion of an object travels at a certain angle.

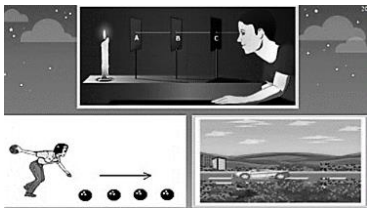


*Figure SEQ Figure * ARABIC 1.* Examples of Rectilinear Motion

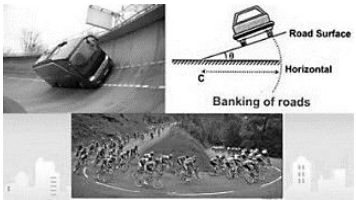


Figure 2. Examples of Curvilinear Motion

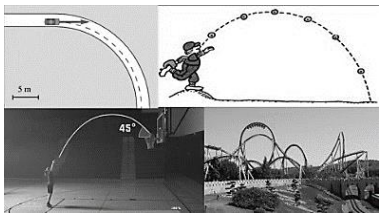
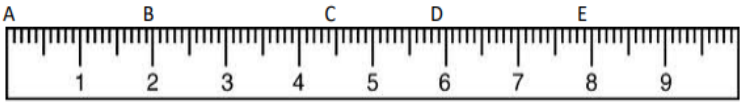


Figure3. Examples of Angular Motion

Can you give examples of bodies in motion?

REFERENCE FRAMES - When making measurements related to motion a frame of reference is needed. This is often a simple number line fixed to a location but when things are moving it can be challenging to determine. Generally, this is a simple x or y axis but if you are on the train, you might say the world is moving relative to you and the train.

POSITION - An object's position can be defined by comparing it to a ruler or some other measuring device. A number may be assigned to it. (The longer lines are centimeters; the shorter ones are millimeters.



What is the position of? A_____ B _____ C _____ D_____ E_____

Motion is mathematically described in terms of displacement, distance, velocity, acceleration, speed, and time.

Distance vs. Displacement

- Distance is the total length of path travelled by an object regardless of the direction. It is a scalar quantity.
- Displacement is the actual measure of the difference between the initial position up to the final position. It is a vector quantity.

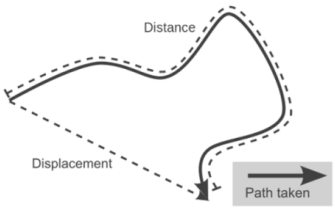


Figure 4. Distance vs. Displacement

Example: A person walks 4 m to the North, 3 m East, 4 m South, and finally, 3 m West. What is the distance travelled by the person? What was his displacement?
Answer: distance = 14 m; displacement = 0

Speed vs. Velocity

- Speed is the measure of how fast something is moving. It is the rate at which distance is covered at a given time. It is a scalar quantity. (speed/time)
- Velocity is the speed at which something moves in one direction. It is a vector quantity.
- Instantaneous speed is the speed which is measured at a particular instant.
- Average Speed is the distance travelled divided by the total time elapsed in travelling the distance.
- Average Velocity is the total displacement of an object per unit of time.

Example: A car travels a distance of 540km to the north in 6 hours. Compute for the speed and the velocity.

$$speed = \frac{distance}{time} = \frac{540\ km}{6\ hours} = 90\ km/hr$$

Since velocity is equal to speed + direction, its velocity is 90 km/hr, north

Acceleration

Acceleration is the change in velocity in a particular time interval. An object is accelerating when: (a) it speeds up, (b) it slows down (deceleration), (c) it changes direction.

$$a = \frac{v_2 - v_1}{t}$$

Where: a – acceleration
v_2 – final velocity
v_1 – initial velocity
t - time

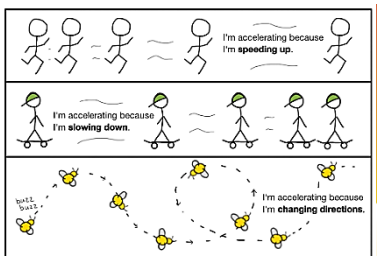


Figure 6. Illustration of acceleration

Example:

A skater increases her velocity from 2.0 m/s to 10.0 m/s in 3.0 seconds. What is the skater's acceleration?

$$a = \frac{v_2 - v_1}{t} = \frac{10\frac{m}{s} - 2\ m/s}{3s} = 2.67\ m/s^2$$

What's More

Directions: Solve the following problems. Use an extra sheet for your complete solution.

Boots walks 100 m West on his way to market but return back at a distance of 25 m East after he found out that his wallet was lost. What is the distance and displacement?

Neil travelled 36km at a speed of 8km/h. Grant travelled 48km at a speed of 10km/h
a) Whose journey was quickest?
b) By how many mins?

What I Have Learned

Reflection. Explain the question below.

Is it possible for distance to be equal to zero? How about displacement?

_____

_____

What I Can Do

While traveling along a highway a driver slows from 24 m/s to 15 m/s in 12 seconds. What is the automobile's acceleration? (Remember that a negative value indicates a slowing down or deceleration.)

Assessment

Directions: Read each statement carefully. Encircle the letter of the correct answer.

1. Shona cycles at an average speed of 8km/h. How far has she travelled if she cycles for 4 hours?
   A. 55 km B. 43 km C. 32 km D. 22 km
2. If a cheetah runs 180 miles in 3 hours, what is its average speed?
   A. 60 hours B. 60 miles/hour C. 60 miles D. 60 mi/s^2
3. Which of the following is a vector quantity?
   A. Time B. Speed C. Velocity D. Distance
4. A car at accelerates from 0 m/s to 25 m/s in 5 seconds. What is the acceleration of the car?
   A. 30 m/s^2 B. 50 m/s^2 C. 125 m/s^2 D. 5 m/s^2
5. What is the speed of a bobsled whose distance-time graph indicates that it travelled 100 m in 25 seconds?
   A. 4 m/s B. 250 m/s C. 0.25 m/s D. 100 m/s

Lesson 2 Motion Graphs

The study of 1-dimensional kinematics has been concerned with the multiple means by which the motion of objects can be represented. The motion of an object can be represented through the use of words, diagrams, numbers, equations, and graphs. In this module, we will focus on the use of position vs. time graphs to describe motion. As we will learn, the specific features of the motion of objects are demonstrated by the shape and the slope of the lines on a position vs. time graph. The first part of this lesson involves a study of the relationship between the shape of a p-t graph and the motion of the object.

What's In

In Physics, we often use these graphs to present objects in motion. These graphs are called **motion graphs**. As we go along, you will know more about these motion graphs.

Motion graphs have a considerable importance in Physics. Motion graphs are used in describing the motion of objects. In this module, you will learn how to read and interpret various motion graphs.

What is It

What does the vertical axis represent on a position graph?

The vertical axis represents the position of the object. For example, if you read the value of the graph on the right at a particular time, you will get the position of the object in meters. Meanwhile, the horizontal axis represents the time.

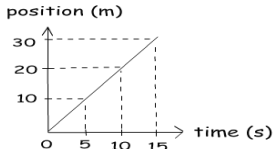
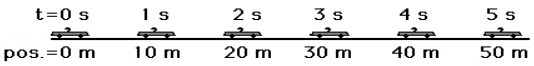


Figure 1. Position vs. time

Concept check: What is the position of the object at time t = 5 seconds according to the graph? Yes, 10 meters. You are correct!



Consider a car moving with a constant, rightward (+) velocity - say of +10 m/s.

If the position-time data for such a car were graphed, then the resulting graph would look like the graph at the right. Note that a motion described as a constant, positive velocity results in a line of constant and positive slope when plotted as a position-time graph.

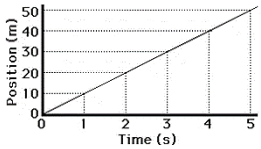
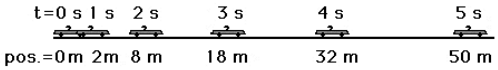


Figure 2. Position vs. time graph of the car with constant velocity



Now consider a car moving with a rightward (+), changing velocity - that is, a car that is moving rightward but speeding up or accelerating.

If the position-time data for such a car were graphed, then the resulting graph would look like the graph at the right. Note that a motion described as a changing, positive velocity results in a line of changing and positive slope when plotted as a position-time graph.

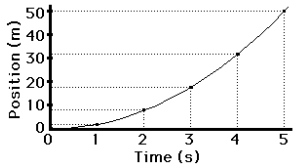
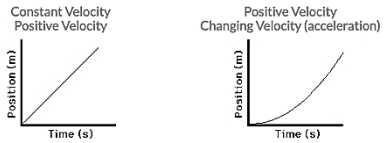


Figure 3. Position vs. time graph of the car with changing velocity

The position vs. time graphs for the two types of motion - constant velocity and changing velocity (acceleration) - are depicted as follows.



What does the slope represent on a position graph?

The shapes of the position versus time graphs for these two basic types of motion - constant velocity motion and accelerated motion (i.e., changing velocity) - reveal an important principle. The principle is that the slope of the line on a position-time graph reveals useful information about the velocity of the object. It is often said, "As the slope goes, so goes the velocity." Whatever characteristics the velocity has, the slope will exhibit the same (and vice versa). If the velocity is constant, then the slope is constant (i.e., a straight line). If the velocity is changing, then the slope is changing (i.e., a curved line). If the velocity is positive, then the slope is positive (i.e., moving upwards and to the right). This very principle can be extended to any motion conceivable.

The slope of a position graph represents the velocity of the object. So, the value of the slope at a particular time represents the velocity of the object at that instant. The slope of this position graph is $slope = \frac{rise}{run} = \frac{x_2 - x_1}{t_2 - t_1}$

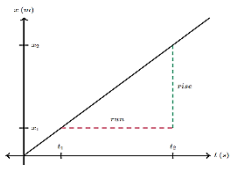
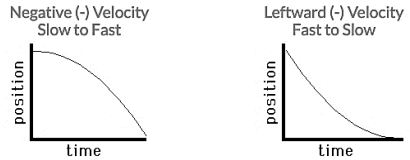


Figure 4. Graph showing the slope

Representing an Accelerated Motion



As a final application of this principle of slope, consider the two graphs on the left. Both graphs show plotted points forming a curved line. Curved lines have changing slope; they may start with a very small slope and begin curving sharply (either upwards or downwards) towards a large slope. In either case, the curved line of changing slope is a sign of accelerated motion (i.e., changing velocity). Applying the principle of slope to the graph on the left, one would conclude that the object depicted by the graph is moving with a negative velocity (since the slope is negative). Furthermore, the object is starting with a small velocity (the slope starts out with a small slope) and finishes with a large velocity (the slope becomes large). That would mean that this object is moving in the negative direction and speeding up (the small velocity turns into a larger velocity). This is an example of negative acceleration - moving in the negative direction and speeding up. The graph on the right also depicts an object with negative velocity (since there is a negative slope). The object begins with a high velocity (the slope is initially large) and finishes with a small velocity (since the slope becomes smaller). So, this object is moving in the negative direction and slowing down. This is an example of positive acceleration.

What's More

Graph and label the scenario to a sketch position vs. time and velocity vs. time. If specific distances, velocities, and times are given, for simplicity's sake, assume all movement is in a straight line.

A car is driving down the road at 55 mph, and after getting a flat tire moves at 35 mph.

What I Have Learned

_____

What I Can Do

How will you determine her total displacement if your mother walks five meters to the kitchen at a velocity of 2 m/s, pauses by the refrigerator for three seconds, then walks back with a plate full of potato salad at a velocity of 1 m/s?

The motions described in each of the questions takes place at an intersection on a two-lane road with a stop sign in each direction. For each motion, select the correct position versus time graph. For all of the motions, the stop sign is at the position x=0, and east is the positive x-direction.

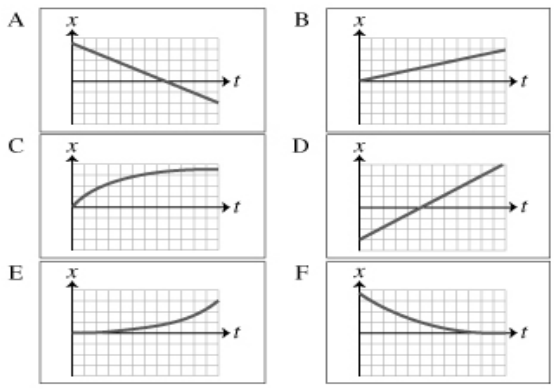
Identify what is being described in each situation.

_____ 1. A driver ignores the stop sign and continues driving east at constant speed.

_____ 2. A driver ignores the stop sign and continues driving west at constant speed.

_____ 3. A driver, traveling west, slows and stops at the stop sign.

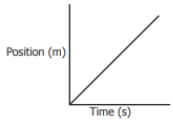
_____ 4. A driver, after stopping at the stop sign, travels east with a positive acceleration.



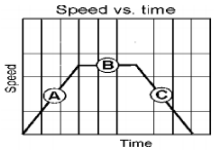
Assessment

Directions: Read each statement carefully. Encircle the letter of the correct answer.

1. What does the following position-time graph tell us about the motion of the object?
   A. The object is accelerating in the positive direction.
   B. The object is moving in the positive direction at a constant velocity.
   C. The object is in uniform motion and headed in the negative direction.
   D. The object is not moving.



For item numbers 2-5, consider the graph on the right.



2. What is happening to the object at point C?
   A. has positive acceleration
   B. has zero acceleration
   C. has negative acceleration
   D. is moving at constant velocity
3. What is happening to the object at point A?
   A. has positive acceleration
   B. has zero acceleration
   C. has negative acceleration
   D. is moving at constant velocity
4. What is happening to the object at point B?
   A. has positive acceleration
   B. has zero acceleration
   C. has negative acceleration
   D. is moving at constant displacement
5. What is the dependent variable?
   A. speed
   B. time
   C. acceleration
   D. distance

Lesson 3 Uniformly Accelerated Motion

We all know that acceleration is the rate at which the velocity of an object changes with respect to time. However, do you know what uniformly accelerated motion is? It can be difficult to perceive how an aspect that is defined by a rate of change can be defined as uniform. Therefore, let us study such uniformity in acceleration in detail, especially since this is a vital concept which is necessary in numerous lessons in physics, too.

What's In

One can say that uniform acceleration definition refers to an acceleration of an object, which remains constant irrespective of time. In simpler terms, a number equal to the acceleration in such a motion does not change as a function of time. Some uniform accelerated motion examples include a ball rolling down a slope, a skydiver jumping out of a plane, a ball dropped from the top of a ladder and a bicycle whose brakes have been engaged. We'll know more about it in this module.

What is It

Uniformly Accelerated Motion (UAM) is motion of an object where the acceleration is constant. In other words, the acceleration remains uniform; the acceleration is equal to a number and that number does not change as a function of time.

Examples of objects in Uniformly Accelerated Motion:

- A ball rolling down an incline.
- A person falling from a plane.

These are the equations that describe an object in Uniformly Accelerated Motion:

$$d = v_i{*}t + \frac{1}{2}{*}a{*}t^2 \qquad v_f^2 = v_i^2 + 2{*}a{*}d \qquad v_f = v_i + a{*}t \qquad d = \frac{v_i + v_f}{2} * t$$

In the above equations, **the symbol d stands for the displacement** of the object. The **symbol t stands for the time** for which the object moved. **The symbol a stand for the acceleration** of the object. And the **symbol v stands for the instantaneous velocity** of the object; **a subscript of i after the v (as in vi) indicates that the velocity value is the initial velocity value and a subscript off (as in vf) indicates that the velocity value is the final velocity value.**

What's More

Directions: Read and answer the problem using the equations in uniformly accelerated motion.

A car starts from rest and accelerates uniformly over a time of 5.21 seconds for a distance of 110 m. Determine the acceleration of the car.

What I Have Learned

An airplane accelerates down a runway at 3.20 m/s2 for 32.8 s until it finally lifts off the ground. Determine the distance travelled before take-off.

What I Can Do

Does uniformly accelerated motion exist in daily life? Justify your answer.

_____

Assessment

Directions: Read each statement carefully. Encircle the letter of the correct answer.

1. At the surface of the Earth where air friction is negligible, objects fall with
   A. different acceleration C. same acceleration
   B. same velocity D. same speed
2. When a ball is thrown straight up with no air resistance, the acceleration at its highest point
   A. is upward C. is zero
   B. is downward D. reverses from upward to downward
3. A package is dropped straight down from the top of a building at a speed of 20 m/s. It hits the ground with a speed of 40 m/s. How high is the building?
   A. 61.2 m B. 62.1 m C. 612 m D. 621 m
4. Starting from rest, a dragster travels a straight 1/4 mi racetrack in 6.70 s with constant acceleration. What is its velocity when it crosses the finish line?
   A. 269 mi/h B.188 mi/h C. 296 mi/h D. 135 mi/h
5. A car initially travelling at 60 km/h accelerates at a constant rate of 2 m/s2. How much time is required for the car to reach a speed of 90 km/h?
   A. 15 s B. 30 s C. 45 s D. 4.2

Kinematics: Motion in 2-3 Dimensions
Quarter 1 – Week 3

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Kinematics: Motion in 2-3 Dimensions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module has one lesson, namely:

- Lesson 1 – Relative Motion, Projectile Motion and Circular Motion

After going through this module, you are expected to:

1. describe motion using the concept of relative velocities in 1D and 2D;
2. define projectile motion;
3. explain the independence of projectile motion in vertical and horizontal components;
4. explain the differences of quantities such as tangential velocity, centripetal acceleration, tangential acceleration, and radius of curvature present in an object in circular motion; and
5. solve problems involving centripetal acceleration of an object moving in circular path.

What I Know

K	W	L
What do I know?	**What do I want to know?**	**What did I learn?**

Directions: Fill in the K-W-L Chart below and with what you Know about the topic, what you Want to know, and what you've Learned to assess your prior knowledge and understanding about motion in 2- or 3-dimensions.

Lesson 1-3 Relative Motion, Projectile Motion and Circular Motion

In rectilinear or straight-line motion, the net force acting on a body is in the direction of motion. If this net force acts at an angle to the direction of motion at any instant, the object will move in a curved path. The arc of basketball, a bicycle rounding a curve and even a cat chasing its tail are a few examples of motions along a curved path. This 2- or 3-dimensional motions are simple extensions of one-dimensional motion in the previous lesson.

What's In

Motion does not happen in isolation. If you are riding a train moving at 10 m/s east, this velocity is measured relative to the ground on which you're traveling. However, if another train passes you at 15 m/s east, your velocity relative to this other train is different from your velocity relative to the ground. Your velocity relative to the other train is 5 m/s west. The motion of an object with respect to other moving or stationary object is called ***relative motion***.

What is It

Relative Motion in One Dimension

We introduce relative motion in one dimension first, because the velocity vectors simplify to having only two possible directions. Take the example of the person sitting in a train moving east. If we choose east as the positive direction and Earth as the reference frame, then we can write the velocity of the train with respect to the Earth as $\vec{v}_{TE}$ = 10 m/s east, where the subscripts TE refers to train and earth. Let's now say the person gets up out of her seat and walks toward the back of the train at 2 m/s. This tells us she has a velocity relative to the reference frame of the train. Since the person is walking west, in the negative direction, we write her velocity with respect to the train as $\vec{v}_{PT}$ = −2 m/s. We can add the two velocity vectors to find the velocity of the person with respect to Earth. This relative velocity is written as: $\vec{v}_{PE}=\vec{v}_{PT}+\vec{v}_{TE}$

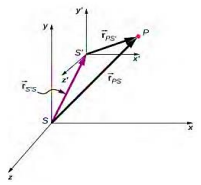
Note the ordering of the subscripts for the various reference frames in Equation 1. The subscripts for the coupling reference frame, which is the train, appear consecutively in the right-hand side of the equation. Figure 1 shows the correct order of subscripts when forming the vector equation.

$$\vec{v}_{PE} = \vec{v}_{PT} + \vec{v}_{TE}$$

Figure 1: When constructing the vector equation, the subscripts for the coupling reference frame appear consecutively on the inside. The subscripts on the left-hand side of the equation are the same as the two outside subscripts on the right-hand side of the equation.

Relative Velocity in Two Dimensions

We can now apply these concepts to describing motion in two dimensions. Consider a particle P and reference frames S and S′, as shown in Figure.3. The position of the origin of S′ as measured in S is $\vec{r}_{S'S}$, the position of P as measured in S′ is $\vec{r}_{PS}$, and the position of P as measured in S is $\vec{r}_{PS}$.



From Figure 3 we see that: $\vec{r}_{PS} = \vec{r}_{PS'} + \vec{r}_{S''S}$

The relative velocities are the time derivatives of the position vectors. Therefore, $\vec{v}_{PS}=\vec{v}_{PS'}+\vec{v}_{S'S}$

The velocity of a particle relative to S is equal to its velocity relative to S′ plus, the velocity of S′ relative to S.

We can extend Equation 3 to any number of reference frames.

particle P relative t S′ are $\vec{r}_{PS}$ and $\vec{r}_{PS}$, For particle P with velocities $\vec{v}_{PA}$, $\vec{v}_{PB}$, and $\vec{v}_{PC}$ in frames A, B, and C,

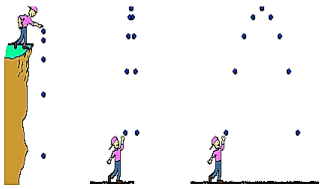
$$\vec{v}_{PC} = \vec{v}_{PA} + \vec{v}_{AB} + \vec{v}_{BC}.$$

Projectile Motion

A cannonball shot from a cannon, a stone thrown into the air, a ball rolling off the edge of a table, a spacecraft circling Earth—all of these are examples of projectiles. A **projectile** is any object that moves through the air or space, acted on only by gravity (and air resistance, if any). It is any object with an initial non-zero, horizontal velocity whose acceleration is due to gravity alone.

There are various types of projectiles (see Figure 1): object dropped from rest, object thrown vertically upward, and object which is thrown upward at an angle to the horizontal.

Figure 1: Types of Projectile



Consider a cannonball projected horizontally by cannon from the top of a very high cliff. In the absence of gravity, the cannonball would continue its horizontal motion at a constant velocity. Furthermore, if merely dropped from rest in the presence of gravity, the cannonball would accelerate downward, gaining speed at a rate of 9.8 m/s every second (this is consistent with our conception of free-falling objects accelerating at a rate known as the **acceleration due to gravity)**.

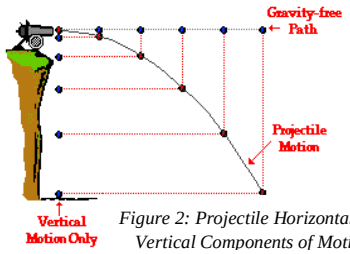


Figure 2: Projectile Horizontal and Vertical Components of Motion

Now, if the cannonball is projected horizontally in the presence of gravity, then the cannonball would maintain the same horizontal motion as before - a constant horizontal velocity. Furthermore, the force of gravity will act upon the cannonball to cause the same vertical motion as before - a downward acceleration. The cannonball falls the same amount of distance as it did when it was merely dropped from rest (refer to figure on the left). However, the presence of gravity does not affect the horizontal motion of the projectile. The force of gravity acts downward and is unable to alter the horizontal motion. The vertical force acts perpendicular to the horizontal motion and will not affect it since *perpendicular components of motion are independent of each other*. Thus, the projectile travels with a *constant horizontal velocity* and a *downward vertical acceleration*. The trajectory of a simple projectile is a *parabola*.

The ***maximum range*** or horizontal distance of a projectile can be reached if launched at 45°, equal ranges for launch angles that exceed and fall short of 45° by equal amounts (ex. 40° & 50°, 30° & 60°, 0° & 90°).

Use the horizontal direction to determine the range (R) as a function of time:

$$x = x_0 + v_{0x}t + \tfrac{1}{2}a_x t^2$$

$$x = 0 + (v\cos\theta)t + 0$$

$$\boldsymbol{x_{final} = (v\cos\theta)\,t_{final}}$$

Equation	Horizontal Component	Vertical Component
Acceleration	$a_x = 0$	$a_y = -g$
velocity-time	$v_x = v_{0x}$	$v_y = v_{0y} - gt$
displacement-time	$x = x_0 + v_{0x}t$	$y = y_0 + v_{0y}t - \tfrac{1}{2}gt^2$
velocity-displacement		$v_y^2 = v_{0y}^2 - 2g(y - y_0)$

Use the vertical direction to determine the time in the air…

$y = y_0 + v_0yt + ½a_yt^2$

$y = y_0 + (v \sin \theta)t - ½gt^2$

$0 = 0 + (v \sin \theta)t_{final} - ½gt^2_{final}$

$$t_{final} = \frac{2(v \sin \theta)}{G}$$

Combine these two equations…

$$x_{final=}(vcos\theta)\frac{2vsin\theta}{g}$$

$$x_{final=}\frac{v^2sin2\theta}{G}$$

$$x_{max} = \frac{v^2sin2\theta}{G}$$

What's More

In physics, **circular motion** is a **movement** of an object along the circumference of a **circle** or rotation along a **circular** path. It can be uniform, with constant angular rate of rotation and constant speed, or non-uniform with a changing rate of rotation.

When an object moves in a circle at a constant speed its velocity (which is a vector) is constantly changing. Its velocity is changing not because the magnitude of the velocity is changing but because its direction is. This constantly changing velocity means that the object is accelerating (**centripetal acceleration**). For this acceleration to happen there must be a resultant force, this force is called the **centripetal force**.

The angular speed of an object moving in a circle is given by the following equation:

$$\omega = \frac{\theta}{t} \qquad \omega = \frac{v}{r} \qquad \omega = 2\pi f \qquad \omega = \frac{2\pi r}{T}$$

where ω is the angular speed, v is the linear speed, θ is it angle it moves through measured in **radians (rad),** t is the time, r is the radius and f is the frequency.

The angular acceleration of an object moving in a circle can be determined by either two of the following equations: $\alpha = \frac{v^2}{r}$ $\qquad \alpha = \omega^2 r$ $\qquad \alpha = \frac{4\pi^2 r}{T^2}$

where v is the speed, a is the acceleration, r is the radius, and T is the period. The equation on the right (above) is derived from the equation on the left by the substitution of the expression for speed.

In rotational motion, **tangential acceleration** is a measure of how quickly a **tangential** velocity change. It always acts perpendicular to the centripetal **acceleration** of a rotating object. It is equal to the angular **acceleration** α, times the radius of the rotation. Tangential acceleration is a measure of how quickly a tangential velocity change. It always acts perpendicular to the centripetal acceleration of a rotating object. It is equal to the angular acceleration α, times the radius of the rotation. In symbols: $\boldsymbol{a_{tan} = ra.}$, where a_{tan} is the tangential acceleration, r is the radius of the object's rotation and a = angular acceleration, with unit radians/s^2

The centripetal force ($\mathbf{F_{net}}$) acting upon an object moving in circular motion is directed inwards. While there may by more than one force acting upon the object, the vector sum of all of them should add up to the net force. In general, the inward force is larger than the outward force (if any) such that the outward force cancels and the unbalanced force is in the direction of the center of the circle. The centripetal force, measured in Newtons (N) can be different forces in different settings it can be gravity, friction, tension, lift, electrostatic attraction etc. The net force is related to the acceleration of the object (as is always the case) and is thus given by the following three equations:

$$F_{net} = ma \qquad F_{net} = m\frac{v^2}{r} \qquad F_{net} = m\,\omega^2 r \qquad F_{net} = m\frac{4\pi^2 r}{T^2}$$

The equations for net force in the middle and on the right are derived from the equation on the left by the substitution of the expressions for acceleration.

Sample Problem: A 900-kg car moving at 10 m/s takes a turn around a circle with a radius of 25.0 m. Determine the acceleration and the net force acting upon the car.
The solution of this problem begins with the identification of the known and requested information.
Given: m = 900 kg, v = 10.0 m/s, r = 25.0 m *Find*: a =? F_{net} =?
Use equation a = v^2 / r.

$$a = \frac{v^2}{r} = \frac{(100m/s)^2}{25.0\,m} = \mathbf{4\ m/s^2}$$

To determine the net force acting upon the car, use the equation F_{net} = m•a. The solution is as follows. F_{net} = ma = (900 kg) (4 m/s^2) **= 3600 N**

What I Have Learned

Compare relative motion in one dimension to relative motion in two dimensions.

What I Can Do

Directions: Read and solve the following problems. Write all pertinent solutions in another answer sheet.

1. Two cars, standing a distance apart, start moving towards each other at the speed of 1 m/s and 2 m/s along a straight road. What is the speed when they approach?
2. A soccer ball is kicked horizontally off a 22.0-m high hill and lands a distance of 35.0 meters from the edge of the hill. Determine the initial horizontal velocity of the soccer ball.
3. A race car makes one lap around a track of radius 50 m in 9.0 s. a) What is the average velocity? b) What was the car's centripetal acceleration?

Assessment

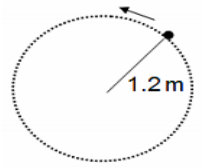
Directions: Choose the letter of the correct answer. Write the letter of your choice on the space provided before each number. (USE CAPITAL LETTER ONLY)

___1. A duck waddles 2.5 m east and 6.0 m north. What are the magnitude and direction of the duck's displacement with respect to its original position?
A. 3.5 m at 19° north of east C. 6.5 m at 72° north of east
B. 6.3 m at 67° north of east D. 6.5 m at 67° north of east

___2. A track star in the long jump goes into the jump at 12 m/s and launches herself at 20.0° above the horizontal. How long is she in the air before returning to Earth? (g = 9.8 m/s^2)
A. 1.5 s B. 1.2 s C. 0.42 s D. 0.83 s

___3. A model rocket flies horizontally off the edge of the cliff at a velocity of 50.0 m/s. If the canyon below is 100.0 m deep, how far from the edge of the cliff does the model rocket land?
A. 112 m B. 337 m C. 225 m D. 400 m

For question nos. 4-5: A 0.2 kg ball rotates at a constant speed of 3 m/s on the end of 1.2 m long string. (Refer to the diagram on the right) The string describes a horizontal circle.



___4. What is the centripetal acceleration of the object?
A. 1.2 m/s^2 B. 3.0 m/s^2 C. 7.5 m/s^2 D. 3.2 m/s^2

___5. What is the centripetal force exerted on the object?
A. 1.0 N B. 1.2 N C. 0.2 N D. 1.5 N

Newton's First Law of Motion
Quarter 1 – Week 4

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Newton's First Law of Motion. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.
The module has two lessons, namely:

- Lesson 1 – Forces and Free-Body Diagram
- Lesson 2 – Newton's First Law of Motion

After going through this module, you are expected to:

1. define inertial frames of reference;
2. draw free-body diagrams;
3. apply Newton's 1st law to differentiate contact forces and noncontact forces;
4. illustrate examples of contact and noncontact forces; and
5. differentiate the properties of static friction and kinetic friction.

What I Know

Directions: Answer the questions below on a separate sheet of paper:

1. What properties do forces have that allows us to classify them as vectors?
2. How are inertia and mass related?
3. What is the relationship between weight and mass?

Lesson 1-2 Forces and Free-Body Diagram

By defining position, speed, velocity, and acceleration, **kinematics** provides us description on how fast or slow a body is moving. However, it does not tell us *why* the body is behaving the way it is. This is why **dynamics** is important. It gives us knowledge about the causes of motion. These two studies make up **mechanics –** the general science of motion.

What's In

How can we apply forces to massive objects like cars?
What force is present when you walk around the classroom?

What is It

As you kick a stone, push the table, and even play tug-of-war, forces are applied. We can say that some sort of *force* is needed to move objects. But *what really is a force*?

1. A force is a push or a pull. It is applied in everyday language. However, a better definition is that a force is an interaction between two bodies or between a body and its environment.
2. A force is a vector quantity. It has both magnitude and direction.
3. A force requires an agent. Something does the pushing or pulling.
4. A force is either a contact or non-contact (long range) force. We will focus more on gravity as we deal with long-range force.

The SI unit of force is 1 Newton (1**N**) which is equal to 1 kg $\frac{m}{s^2}$.

What's More

Kinds of Forces

As we go through our discussion in dynamics, we will be dealing with different kinds of forces. It is very important to familiarize yourself on each:

Weight - Gravity plays a big role in the structure of universe. The gravitational pull of the earth on an object on or near the surface of the earth. The symbol for weight is *w*. It is equal to the product of mass (*m*) of the object and the gravitational acceleration (*g*), expressed as (*w=mg*).
Spring Force - Springs is one of the most common contact forces. A spring can either push (when compressed) or pull (when stretched).
Tension Force - When a string or a rope pulls on an object, it exerts a contact force on it, which is represented by a *T*. The direction of the force is always on the direction of the rope. When you tug a dog's leash, the force that pulls on its collar is also a tension force.
Normal Force The force exerted perpendicularly by a surface against an object pressing on that surface. A book lying on a table experiences a normal force. As you sit on a chair, the chair exerts an upward force on you. The normal force is responsible for the 'solidness' of solids. It keeps you from passing right through the chair you are sitting in.
Friction Force - Most objects stay in place on a table without sliding off. These things happen because of the presence of **friction force.** Like a normal force, it is also exerted by a surface. But it is important to take note the two kinds of friction force which are also presented in Figure 1.:

Figure 1. Kinetic and Static Friction
Physics for Scientists and Engineers with Modern Physics: A Strategic Approach (p. 103).

Kinetic Friction (f_k) is the force causes an object slides across a surface. This is the kind of force which 'opposes the motion' of the body, meaning that the friction force vector f_k points in a direction opposite to the motion.

Static Friction (f_s) is the force that keeps an object 'stuck' on a surface which also prevents its motion. Static friction points opposite the direction in which the body would move if friction is not present.

Properties of Friction

1. Friction always acts parallel to two planes in contact with each other and in the direction opposite the motion.
2. Friction always decreases the relative motion between the bodies and the surface.
3. Friction is greater in rough surfaces than in smooth surfaces.

Free-Body Diagram

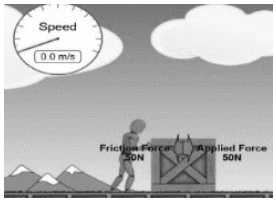
Newton's Laws of Motion give us an ability to solve problems in mechanics
A **Free-Body Diagram** represents the body as a particle which shows all the forces interacting on it. The length of the line of an arrow represents the magnitude of the force while its tip represents the force's direction. "How are forces presented in a single diagram?"

Drawing a Free Body Diagram:
Identify all the forces acting on the object.
Draw a coordinate system in accordance to the orientation of the body. If the body is tilted, the orientation of the coordinate system has to be tilted as well.
Represent the object as a dot at the origin of the coordinate axes.
Draw vectors representing each force. Draw the arrow in accordance to the direction and magnitude of the force.
Draw and label the *net force vector* F_{net}. Draw this vector beside the diagram, not on the particle.

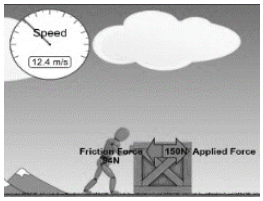
Activity: Push Me Hard

Directions: Shown below are pictures from a simulation in which friction is applied (If you have an internet connection, you may see the simulation by scanning the QR code below or using this link: https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html): Then answer the guide questions below.

A



B



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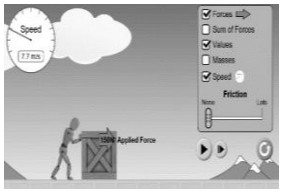
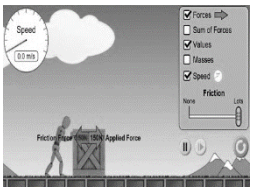


D



Guide Questions:

1. On all diagrams, where is the direction of the friction force?
2. On diagram A, the man pushed the crate by applying 50 N of force on it. Did the crate move? Why or why not?
3. What kind of friction is applicable on diagram A? Why do you say so?
4. By looking at diagram B, the man increased his exerted force from 50 N (shown in diagram A) to 150 N. Did the crate (still) move? Why or why not?
5. What kind of friction is seen on diagram B? Why?
6. Comparing diagram C and diagram D, what is the difference having no friction and having lots of friction experienced by the body?

What I Have Learned

Given the scenario below, make a free-body diagram of the forces experienced by the car on the box provided by following the procedures that follow:

Source: Siyavula Uploaders. (n.d.). [A man pushing a car]. Retrieved from https://cnx.org/contents/PCYjGBH3@1/To-identify-the-effect-of-forces

1. List down all the forces experienced by the car. Make a coordinate system with respect to the orientation of the car. Represent the car as a point placed on the origin.
2. Assuming that the car moves, draw each force as vectors on the coordinate system (be mindful on the length of each vector) using the different colors of the pen.
3. Identify the direction of the net force.

What I Can Do

What are the advantages and disadvantages of friction? How are you going to increase/decrease the friction experienced by a body? How important are free-body diagrams in solving problems in mechanics?

_____

_____

Assessment

Directions: Read each item carefully. Write **T** if the statement is true, and **F** if otherwise.
___1. Friction occurs only if the object is moving.
___2. Friction always opposes the motion of the body.
___3. Friction is greater in rough surfaces than in smooth surfaces.
___4. Friction is perpendicular to the direction of the motion of the body.
___5. Friction is a non-contact force.

Lesson 3 Newton's First Law of Motion

Force in simple word is a strength use in physical action. On the other hand, force in science is simply a push or a pull to an object. You can apply this force with or without touching each other and can cause objects at rest to accelerate.

What's In

How can we apply forces to massive objects like cars?
What force is present when you walk around the classroom?
Is there force present when we comb our hair with plastic comb?

What is It

Do forces really affect the motion of an object? Let us start by defining *zero* net force. We can clearly say that if an object is not in motion, or stopped, it has *zero* net force. But how about objects that are in motion?

As you roll a marble on the floor, it continues to roll until such time that it stops. This is because of the presence of friction. But what if the surface is frictionless? What do you think will happen? If you think that it will continue to move, you are correct! This is because no friction is present to oppose the motion of the marble. With this, we can say that force is not required to *sustain* motion.

What's More

Newton's First Law of Motion states that *"an object at rest stays at rest, while bodies in motion remains in motion travelling at constant velocity in a straight path if and only if the net force acting on the body is zero"*.

Since the body is moving at constant speed, we can say that it is not accelerating. Therefore, the net force is zero.

An object on which net force is zero is said to be in ***mechanical equilibrium***. According to Newton's First Law, there are two types of mechanical equilibrium:

1. ***Static equilibrium*** takes place when the object is at rest.
2. ***Dynamic equilibrium***, on the other hand, takes place when the object is moving in a **straight path** with **constant**.

Newton's First Law gives us a more complete definition of force. If an 'influence' on an object causes it to change its velocity, this influence is the force.

Inertia is the tendency of a body to resist sudden change in motion. All objects have the ability to resist changes in their motion. Its capability to either continue to move or stay at rest depends on this property. Suppose you have a less massive chair and a more massive table. You push both of them at the same time, exerting equal amount of force. The table has more ability to

resist the force you exert than the chair. With this, we can conclude that the greater the mass, the greater the inertia.

As a jeepney suddenly stops, passengers are somehow moving forward near the front seat. Passengers have a real forward acceleration *with respect to the jeepney,* is there a force pushing them forward?

Inertial Frame of Reference is a coordinate system in which Newton's Laws of Motion are valid. The first law provides a convenient way to see if a coordinate system is a frame of reference. Not all coordinate systems are frames of reference. As seen on Figure 2 (a), a ball was placed on the floor of a plane having a constant velocity (zero acceleration). Having no horizontal force, the ball stays on its original position and remains at rest *with respect to the airplane*. With these conditions, we can say that the Newton's First Law is satisfied, which makes the airplane an inertial frame of reference. On Figure 2 (b), as the airplane accelerates, the ball rolled towards the back. Nothing exerts a horizontal force on the ball. However, the ball accelerated *with respect to the airplane*. This scenario violates Newton's First Law of Motion which means that the airplane is *not an inertial frame of reference.* With this, we can say that accelerating reference frames are not inertial reference frames where Newton's Laws of Motion are not valid.

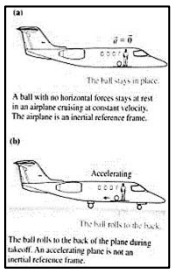


Figure 2. Inertial Frame of Reference
Pearson Education, Inc. (2004c). Reference Frames [Illustration]. In *Physics for Scientists and Engineers with Modern Physics: A Strategic Approach* (p. 112).

What I Have Learned

Directions: Write T if the statement is true, and F if otherwise.

_____1. Newton's First Law of Motion applies in an inertial frame of reference.
_____ 2. An object under the Law of Inertia has zero net force.
_____ 3. The lesser the mass, the greater the inertia.
_____ 4. There are no forces acting on the body if its net force is zero.
_____ 5. Newton's First Law of Motion states that a body will be on its original motion unless acted upon by an external force

What I Can Do

Activity: Yank Me

Materials:

4 plastic cups, 3 index cards (size should cover the area of the plastic cups)

Procedures:

1. Stack the plastic cups and index cards alternately.
2. Pull each cardboard away from the tower without holding the plastic cups.
3. Do these steps from top to bottom.
4. Note that the tower should not fall as you pull each index cards

SCAN ME

(You may watch the process on this task by scanning the QR code on the left or by opening this link: https://www.youtube.com/watch?v=gEQxQPdbZik)

Guide Questions*:*

1. What happens when the index card and plastic cups stay at rest?
2. What happened as you move the index card away from the stack with plastic cups? Did it fall? What could be the possible reason behind it?
3. If we place the set – up inside a moving transportation with a constant velocity, can we say that the transportation is an inertial frame of reference? Why?
4. If we place the set – up inside an accelerating vehicle, can we say that the transportation is an inertial frame of reference? Why?

Assessment

Directions: Essay: On a separate sheet, answer the following questions that follow:

1. Is it possible for an object to move in a constant velocity upon experiencing a force? Why or why not?
2. Does zero net force imply that there are no forces acting on the object?

Newton's Second Law of Motion
Quarter 1 – Week 5

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Newton's Second Law of Motion. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.
The module has three lessons, namely:

- Lesson 1 – Newton's Second Law of Motion
- Lesson 2 – Newton's Law of Motion: Problem Solving
- Lesson 3 – Scalar Product and Work done by a force

After going through this module, you are expected to:

1. solve problems using Newton's Laws of motion;
2. calculate the dot or scalar product of vectors; and
3. determine the work done by a force acting on a system.

What I Know

Directions: Fill in the blanks with the correct word or phrase.

1. Newton's second law of motion is also known as the LAW OF _____.
2. Newton's second law says that when an _____ force is applied to a _____, it causes it to _____.
3. The greater the force that is applied, the _____ the acceleration.
4. If the same force is applied to an object with a small mass, it will have _____ acceleration.
5. An object with a mass of 5.0 kg has a force of 20.0 newtons applied to it. What is the resulting acceleration of the object? _____

Lesson 1 Newton's Second Law of Motion

What makes objects accelerate? Display a block of foam and a brick; ask "Which do I have push to harder, or apply more force, to make it move?" "Which do you think will go fastest?" In this lesson, we will learn the relationship of force, mass and acceleration as stated in Newton's second law of motion.

What's In

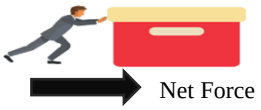
What is the relationship of force to acceleration?
What is the relationship of force to mass?
What is the relationship of mass to acceleration?

What is It

According to Newton, an object will only accelerate if there is a ***net force*** acting upon it. The presence of a net force caused by the unbalanced forces will accelerate an object - changing

its speed, its direction, or both its speed and direction. Simply put, the net force will cause a change in the state of motion of an object.

(a) *A man pushing a stationary box into motion.*



(b) *A tug of war ending in the two people pulling towards the left.*

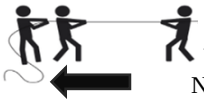


Fig. 1 Presence of an Unbalanced Forces

(You may review the concept about unbalanced and balanced forces by scanning the QR code on the left or by opening this link: https://www.physicsclassroom.com/Concept-Builders/Newtons-Laws/Balanced-Unbalanced-Forces/Interactive)

SCAN ME

Newton's second law of motion can be formally stated as follows: *The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force*, in the same direction as the net force, and inversely proportional to the mass of the object.

In equation form: $a = \frac{F_{Net}}{m}$ and can be rearranged as, $\mathbf{F_{net} = m \bullet a}$

The unit for force is equal to 1 N (Newton), which can be derived by substituting the SI units from mass and acceleration. 1 N = 1 kg • m/s^2

What's More

Example

Your sister shoves a bottle with a mass of 0.78 kg to the right side of a table. The bottle leaves her hand moving at 1.3 m/s, then slows down to a stop due to horizontal friction. If the bottle moved 1.8 m before coming to rest, find the magnitude and direction of the friction force acting on the bottle.

Given

$v_x = 0$ m = 0.78 kg $v_{0x} = 1.31$ m/s x = 1.81 m

Solution

$v_x = v_{0x}^2 + 2a_x(x-x_0)$, but $v_x = 0$ since the bottle came to a stop

$a_x = \frac{-v_{0x}^2}{2(x-x_0)} = -0.47$ m/s^2

Using a_x to find *f* (friction force)

$\sum F_x = -f = ma$ (only friction force acts on the bottle along the x-axis)

$-f = ma$ (the negative sign indicates the **direction** towards the *left* since friction acts opposite to an object's motion)

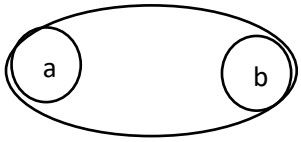
$-f = (0.78 \text{ kg})(-0.47 \text{ m/s}^2) = \mathbf{-0.37\ N}$

Activity: Release Me

Materials: 3 Rubber band chain, empty water bottle, water

Part A.

1. Use 1 rubber band chain make a stretched loop (Fig. 2)
2. **Object a** will be the empty water bottle and object b can be any sturdy stationary object such as a leg of a chair or a pole. Position the band at the center of the bottle.



3. Release your hold on the rubber band, and record your observation.
4. Fill the bottle with water up to 50% capacity, then repeat step 3.
5. Fill the bottle to the brim, then repeat step 3.
6. Adjust the rubber band by adding 2 chains to the loop, then repeat step 3.

Guide Questions:

1. In the experiment, which acts as the unbalanced force? Why?
2. Which setup caused the greatest acceleration, and why?
3. Compare the bottle's acceleration when it was empty and when it was filled with water.

4. Imagine if the bottle upon release, doesn't collide with object b, but slows down as it comes to rest. As it was slowing down, draw a diagram to indicate the direction. Does the velocity, acceleration and force act?

Part B.

Solve the following:

1. What magnitude of net force is required to give a 135 kg man an acceleration of 1.40 m/s^2? _____

2. A 90 g toy car undergoes an acceleration in a straight line (x-axis). The graph on Fig. 2 shows the a-t graph.

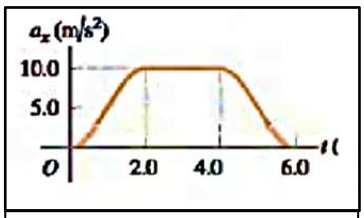


Fig. 2 Acceleration vs time graph
Source. University Physics with Modern Physics

   a. Find the maximum net force, and when does it occur
   b. At what time is the net force on the car constant?
   c. When is the net force exerted on the car zero?

3. You apply a constant horizontal force of 80.0 N to a wooden crate on a smooth horizontal floor (frictional force is negligible). The crate starts from rest and moves 11.0 m in 5.00 seconds.
   a. Determine the mass of the crate.
   b. If you stop pushing at the end of 5.00 s, how far does the crate move in the next 5.00 s?

What I Have Learned

What can you say on the relationship of mass and net force to an acceleration of a certain body?

_____

_____

What I Can Do

Directions: State your ideas on the following scenario of Newton's Second law of motion.

1. If the net force acting on a sliding object increased by a factor of 4, how would it affect the acceleration? _____
2. How does the direction of acceleration compare with the direction of the net force that produces it? _____
3. If the mass of a sliding object is doubled at the same time as the net force is also doubled, how does the resulting acceleration compare with the original acceleration? _____

Assessment

Directions: Read the questions below and answer what is being asked. Show your complete solution.

1. On a long alley, a bowling ball slows down as it rolls. Is any horizontal force acting on the ball? How do you know?
2. A 1750 kg ford mustang is traveling along a track at a constant velocity of 56 m/s. What is the net force acting on the car?
3. A 32.5 kg block of ice is initially at rest, when acted upon a horizontal force of 140 N directed to the right. (a) What is the acceleration of the block of ice? (b) How far does the block travel in 10.0 s? (c)What is the speed at the end of 10.0 s?

Lesson 2 Newton's Law of Motion: Problem Solving

In the world of introductory physics, Newton's second law is one of the most important laws you'll learn. It's used in almost every chapter of every physics textbook, so it's important to master this law as soon as possible. We know objects can only accelerate if there are forces on the object. Newton's second law tells us exactly how much an object will accelerate for a given net force.

If the problem you're analyzing has many forces in many directions, it's often easier to analyze each direction independently.

What's In

How do we use Newton's second law?
What do we do when a force is directed at an angle?
What do solved examples involving Newton's second law look like?

What is It

In solving problems with Newton's law of Motion, one must recall the *summation of forces*.

Sample Problem 1

A hockey puck having a mass of 0.30 kg slides on the horizontal, frictionless surface of an ice rink. Two hockey sticks strike the puck simultaneously (Fig. 1), exerting the forces on the puck shown in Figure 5.4. The force F_1 has a magnitude of 5.0 N, and the force F_2 has a magnitude of 8.0 N. Determine both the magnitude and the direction of the puck's acceleration.

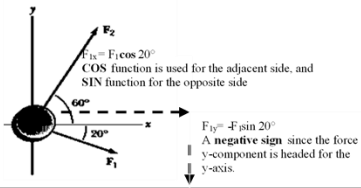


Fig. 1 The x and y components of F_1
Source. Source Serway, R. A., Jewett, J. W., & Serway, R. A. (2004). *Physics for scientists and engineers*. Belmont, CA: Thomson-Brooks/Cole.

Solution

The net force in the x-axis

$\Sigma F_x = F_{1x} + F_{2x}$
$= F_1 \cos(20°) + F_2 \cos 60°$
$= (5.0\ N)\ (0.94) + (8.0\ N0\ (0.5)$
$= \boldsymbol{8.7\ N}$

The net force in the y-axis

$\Sigma F_y = -F_{1y} + F_{2y}$
$= -F_1 \sin(20°) + F_2 \sin 60°$
$= -(5.0\ N)\ (\ 0.342) + (8.0\ N)\ (0.866)$
$= \mathbf{5.2\ N}$

Sample Problem 2

A 2.0 kg steel lunchbox of on a steel horizontal surface is connected to a wooden box of 1.2 kg mass by a lightweight cord over a lightweight, frictionless pulley, as shown in *Fig. 2*. A force of 5.63 N at an angle with the horizontal is applied to the block as shown. The coefficient of kinetic friction between the box and surface is 0.57. Determine the magnitude of the acceleration of the two objects.

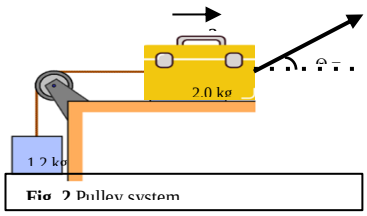


Fig. 2 Pulley system

Given

$m_1 = 2.0$ kg $\quad m_2 = 1.2$ kg $\quad F = 46.6$ N $\quad \mu_k = 0.57$

Solution

Motion of Steel Lunchbox

$\sum \mathbf{F_x} = F\cos\Theta - f_k - T = m_1 a_x = m_1 a$ (since the objects are connected all their acceleration are the same. $a_x = a_y = a$)

$\mathbf{T} = F\cos\Theta - f_k - m_1 a$

$\sum \mathbf{F_y} = n + F\sin\Theta - m_1 g = 0$

$\mathbf{n} = m_1 g - F\sin\Theta$

Motion of Wooden Box

$\sum \mathbf{F_y} = T - m_2g = m_2a_y = m_2a$, (we won't be doing the summation forces along x since there are no forces acting on that axis and, observe no movement along the x-axis)

To solve this, we need to substitute the T from our 3rd equation with the T derived from the 1st equation.

(1) $(F\cos\Theta - f_k - m_1a) - m_2g = m_2a$, Since $f_k = \mu_k n$

(2) $(F\cos\Theta - \mu_k n - m_1a) - m_2g = m_2a$, Substitute n with 2nd equation

(3) $(F\cos\Theta - \mu_k [m_1g - F\sin\Theta] - m_1a) - m_2g = m_2a$

Solve for a

$$\mathbf{a} = \frac{F(\cos\Theta + \mu_k \sin\Theta) - g(m_2 + \mu_k m_1)}{m_1 + m_2}$$

$$\boxed{a = 9.5\ m/s^2}$$

What's More

Activity: Practice more....

Directions: Explain briefly your idea on the following situation. Make a computation if needed.

1. You press your physics textbook flat against a vertical wall with your hand. What is the direction of the friction force exerted by the wall on the book? Justify your answer.
2. Fix the statement "Because the car is at rest, there are no forces acting on it"?
3. Two boxes are being pulled along a frictionless horizontal smooth surface. The boxes are tied to a rope which is being pulled to the right at 270 N. a) Find the acceleration of the system and b) the tension in the rope between the boxes.

What I Have Learned

How do we use Newton's laws of motion in solving problems involving angle?

_____

What I Can Do

Directions: Express your idea on the following condition.

1. As astronaut Yuri performs a routine spacewalk, he notices a large rock floating towards the ship. Should he push it gently or kick it away from the ship? Why?
   Trivia: Contrary to popular belief, The Soviet Union was the first to launch a person in space. America launched their astronaut less than a month later.

2. Can an object exert a force on itself? Argue for your answer. (Clue: 1st law and 2nd law of motion)

Assessment

Directions: Solve the problem, then answer the corresponding question.

1. A 20 kg monkey has a firm hold on a light rope that passes over a frictionless pulley and is attached to a 20 kg bunch of bananas. The monkey sees the bananas, and starts to climb the rope to get them.
   a) As the monkey climbs, do the bananas move up, down, or remain at rest?
   b) As the monkey climbs, does the distance between the monkey and bananas decrease, increase or remains constant?
   c) The monkey releases her hold on the rope. What happens to the distance between the monkey and the bananas while she is falling?
2. A 5.0 kg object placed on a frictionless, horizontal table is connected to a string that passes over a pulley and then is fastened to a hanging 9.0 kg object, as shown in Fig. 6. Find the acceleration of the two objects and the tension in the string.

Lesson 3 Scalar Product and Work done by a force

We have already studied about the addition and subtraction of vectors in the previous lesson. One of the ways in which two vectors can be combined is known as the scalar product. When we calculate the scalar product of two vectors the result, as the name suggests is a scalar, rather than a vector. In this lesson, you will learn how to calculate the scalar product or "dot product" of two vectors.

What's In

Imagine you're sweeping the floor with a broom. You're pushing down on the broom with a force of 50 Newtons, and the broom is at a 50-degree angle with the floor. So, you're pushing partly down and partly forward. If you push the broom 3 meters across the floor, how much work is done? Well, work is a scalar and is equal to force multiplied by displacement, so we use a dot product.

What is It

The ***scalar product*** of any two vectors is defined as a scalar quantity equal to the product of the magnitudes of the two vectors (*A* and *B*) and the cosine of the angle u between them.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

Where did the cos θ come from?

Figure 1 shows vectors *A* and *B* with an angle θ between them. To avoid confusion, rotate your view of the fig.1 so that it aligns vector *A* to the horizontal(x-axis), by doing so, makes it appear that only vector *B* is at an angle. Now find the x component of ***B*** which is ***B cos θ***, and multiply it to ***A*** (since vector A it already aligned with respect to the x-axis), thus arriving at the product ***A(B cos θ).*** The scalar product is commutative, so this can be also expressed as *B* (*A* cos θ).

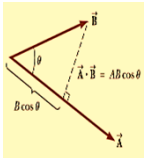


Fig. 1
Source
Serway, R. A., Jewett, J. W., & Serway, R. A. (2004).

The scalar product of two vectors is also the sum of the products of their respective components.

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y$$

Sample Problem

a) Determine the ***scalar product*** *between the two vectors:* A = (4, 3); B = (2, 5)

Given: $A_x = 4$ $B_x = 2$ $A_y = 3$ $B_y = 5$

Solution: $A \cdot B = A_xB_x + A_yB_y = (4 \cdot 2) + (2 \cdot 5) =$ **18**

b) What is the angle between A and B?

Solution

Solve for A and B

$A = \sqrt{[(4)^2 + (3)^2]} = 5$

$B = \sqrt{[(2)^2 + (5)^2]} = 5.39$

Solve for the θ

$\cos \theta = \frac{(A \cdot B)}{AB} = \frac{18}{5(5.39)} = 0.67$

$\theta = \cos^{-1}(0.67) =$ **47.93°**

How can you tell if an object is doing work?

In Fig. 2, you can tell which person achieves work when the one who moves or has *displacement* (d), and those who exert an effort or *force* (F).

Fig. 2 Source www.canva.com

We define ***work*** done on a system by exerting a constant *force* on the system which is the product of the magnitude of the ***force***, the magnitude of the ***displacement*** of the point of application of the force, and ***cos θ***, where θ is the angle between the force and displacement vectors. The unit for work is Joule (J), which is equivalent to 1 J = 1 N·m = 1 kg· m²/s².

$$W = \vec{F}\vec{d}\cos\theta$$

Let's examine the work done on the examples below:

Performing military press

Fig. 3
Source www.canva.com

F_{app} [illegible]
d is directed upward
θ is zero
so, **work done is positive**.

Pushing a wall

Fig. 4
Source www.canva.com

F_{app} is directed to the right
d is zero
so, **work done is zero**.

Push-ups

Fig. 5
Source www.canva.com

F_{app} is directed downward
d is directed upward
θ is 180°
thus, **work done is negative**.

What's More

A. Solve the problem below.
Given the vectors X = (2,3) and Z = (-1,2)
What are the **scalar product** and **angle** between X and Z?

B. Essay: Write and express your idea
Which requires more work: stretching a strong spring a certain distance or carrying your backpack across the hall? Defend your answer. _____

What I Have Learned

How can you say that an object is doing work? _____

What I Can Do

Perform the following situation below.

1. Three children are playing tug-of-war, Tim and Tom against Tam are in a stalemate. Reposition Tim and Tom so that they can win against Tam. **Note: Tim and Tom are found on the right portion of Fig. 5.**
2. Write an explanation. Is more work required to bring a fully loaded bus up to a given speed, than the same but empty bus? Defend your answer.

Fig. 5
Source www.canva.com

Assessment

A. Find the angles on the following questions. Show all your pertinent solutions.
   a) V= (3, -2) and I = (4,-4)
   b) O= (-2, 6) and S = (5,7)
B. Read the problem below and answer the question explicitly on a separate sheet of paper.
   A ball is resting on top of a hill, suddenly it rolls accelerating downward. In this scenario, explain how we can observe simultaneously a negative and positive work done on the ball. Disregard F_{app} and F_{air} in your answer.

Work and Energy
Quarter 1 – Week 6

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topics on work and energy. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module has three lessons, namely:
- Lesson 1 – Work done by a force in one-dimension
- Lesson 2 – Gravitational Potential Energy
- Lesson 3 – Conservative Forces

After going through this module, you are expected to:
1. define work as a scalar or dot product of force and displacement;
2. Relate the gravitational potential energy of a system or object to the configuration of the system; and
3. explain the properties and the effects of conservative forces.

What I Know

Directions: Read each statement carefully. On the space provided before each number, write TRUE if the statement is correct. Otherwise, write FALSE.

_____ 1. Work is a form of energy, and in fact it has units of energy.
_____ 2. Watt is the standard metric unit of power; Joule is the standard metric unit of energy.
_____ 3. _____ is a component of force in the direction of displacement and so this is an example of work.
_____ 4. There is a force and a displacement; the force acts in the opposite direction as the displacement and so this force does negative work.
_____ 5. The upward force does not cause the horizontal displacement so this is a NON-example of work.

Lesson 1 Work done by a force in one-dimension

This module contains lessons and activities on the definition of work as a scalar product of force and displacement as well as the graphical analysis of one-dimensional work done by a force in a Force vs. Position graph. You may answer directly to this activity sheets and make sure to follow the directions stated in each part of the activity. Answer all questions the best that you can and please write eligibly.

What's In

After accomplishing your week 5 module in General Physics 1, you now understand the basic concepts in Physics. You have learned about Newtons's Law and Work. The next phase of your lesson will be the understanding of the concept of Work and its relation to energy and how we analyze graphs. In this module, you are going to apply what you have learned in the first module. Different activities and tasks were prepared to guide you in establishing an understanding of what motion is.

What is It

MATHEMATICAL DEFINITION OF WORK

Mathematically, work, W can be expressed by the following equation.

$$W = F\ d \cos\theta$$

where F is the force, d is the displacement, and the angle θ (theta) is defined as the angle between the force F and the displacement vector d. The work W done on an object by an agent exerting a constant force on the object is the product of the component of the force in the direction of the displacement and the magnitude of the displacement. In the case of work (and also energy), the standard metric unit is the Joule (abbreviated J). One Joule is equivalent to one Newton of force causing a displacement of one meter. In other words,

Joule is the unit of work.

1 Joule = 1 Newton * 1 meter

1 J = 1 N * m

THE SCALAR PRODUCT OF TWO VECTORS – FORCE AND DISPLACEMENT

Because of the way the force and displacement vectors are combined in equation, it is helpful to use a convenient mathematical tool called the scalar product.

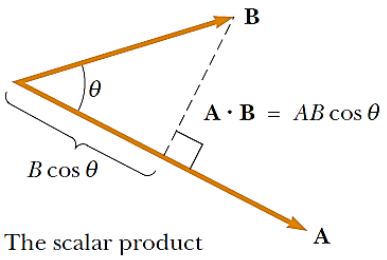
In general, the scalar product of any two vectors A and B is a scalar quantity equal to the product of the magnitudes of the two vectors and the cosine of the angle θ between them:

$$A \cdot B \equiv AB\cos\theta$$

This tool allows us to indicate how F and d interact in a way that depends on how close to parallel they happen to be. We write this scalar product F· d. (Because of the dot symbol, the scalar product is often called the dot product.) Thus, we can express the work equation as a scalar product: **W = F · d = Fd cos**θ

From the right-hand side of the scalar product of two vectors equation we also see that the scalar product is commutative. That is, $A \cdot B = B \cdot A$

The order of the dot product can be reversed



The scalar product **A·B** equals the magnitude of **A** multiplied by $B \cos\theta$, which is the projection of **B** onto **A**.

Finally, the scalar product obeys the distributive law of multiplication, so that $A \cdot (B + C) = A \cdot B + A \cdot C$

The dot product is simple to evaluate from equation when ***A*** is either perpendicular or parallel to ***B***. If ***A*** is perpendicular to ***B***. **(**$\theta = 90°$), then $A \cdot B = 0$. (The equality $A \cdot B = 0$ also holds in the more trivial case when either ***A*** or ***B*** is zero.) If vector ***A*** is parallel to vector ***B*** and the two point in the same direction **(**$\theta = 0°$**),** then $A \cdot B =$ ***AB***. If vector ***A*** is parallel to vector ***B*** but the two point in opposite directions **(**$\theta = 180°$**)**, then $A \cdot B = -AB$. The scalar product is negative when $90° < \theta < 180°$**.**

The unit vectors **i**, **j**, and **k**, which were defined in Week 1, lie in the positive x, y, and z directions, respectively, of a right-handed coordinate system. Therefore, it follows from the definition of $A \cdot B$ that the scalar products of these unit vectors are

$$i \cdot i = j \cdot j = k \cdot k = 1$$
$$i \cdot j = i \cdot k = j \cdot k = 0$$

Therefore, from Week 1, we can state that two vectors ***A*** and ***B*** can be expressed in component vector form as

$$A = Ax\ i + Ay\ j + Az\ k$$
$$B = Bx\ i + By\ j + Bz\ k$$

Using the information given, the scalar product of ***A*** and ***B*** reduces to

$$A \cdot B = Ax\ Bx + Ay\ By + Az\ Bz$$

What's More

Sample Problem:

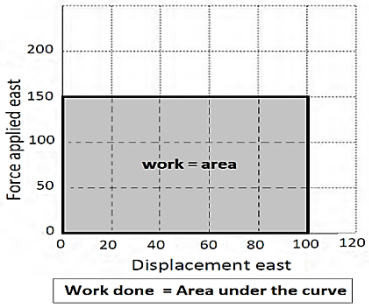
A particle moving in the xy plane undergoes a displacement **d** = (2.0**i** + 3.0**j**) m as a constant force **F**=(5.0**i** + 2.0**j**) N acts on the particle. (a) Calculate the magnitude of the displacement and that of the force. (b) Calculate the work done by **F.**

Solution:

(a) Calculate the magnitude and displacement of the force	(b)) Calculate the work done by **F.**

GRAPHICAL ANALYSIS OF WORK

Work done in a force vs position graph is the area under the curve.



Example

Calculate the work done in the graph on right.

Solution

The shape of the shaded area under the graph is a rectangle. Therefore, work done is equal to the area of the rectangle.

Work = Area of Rectangle

= Force x Displacement

= 150N x 100m

= 15,000 Nm

=15,000 J

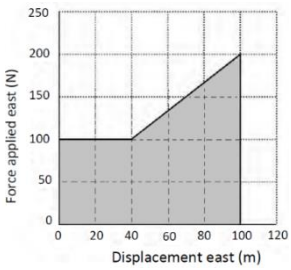
What I Have Learned

How will you operationally describe scalar product of work done? _____

What I Can Do

Directions: Solve the problems below on a separate sheet of paper.

1. Vector A has a magnitude of 5.00 units, and vector B has a magnitude of 9.00 units. The two vectors make an angle of 50.0° with each other. Find $A \cdot B$.
2. Calculate the work done on an object as illustrated by the graph on the right. Hint: Calculate the area or the rectangular section plus the area of the triangular section.



Assessment

Directions: Solve the problem. Use a sheet of paper

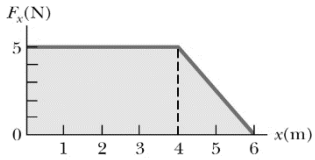
1. **Scalar Product of Force and Displacement.**

   A force **F** = (6.0**i** – 2.0**j**) N acts on a particle that undergoes a displacement **d** = (3.0**i** + **j**) m.

   Find (a) the work done by the force on the particle, (b) calculate the work done and (c) find the angle between F and d.

2. **Calculating Total Work Done from a Graph**

   A force acting on a particle varies with x, as shown in the figure on right. Calculate the work done by the force as the particle moves from x=0 to x=6.0 m.



Lesson 2 Gravitational Potential Energy

In this lesson, we introduce another form of energy—potential energy, which is the energy associated with the arrangement of a system of objects that exert forces on each other. Potential energy can be thought of as stored energy that can either do work or be converted to kinetic energy.

The potential energy concept can be used only when dealing with a special class of forces called *conservative forces*. When only conservative forces act within an isolated system, the kinetic energy gained (or lost) by the system as its members change their relative positions is balanced by an equal loss (or gain) in potential energy. This balancing of the two forms of energy is known as the *principle of conservation of mechanical energy.*

What's In

This worksheet contains lessons and activities on potential energy, which is the energy associated with the arrangement of a system of objects that exert forces on each other. You may answer directly to this activity sheets and make sure to follow the directions stated in each part of the activity. Answer all questions the best that you can and please write eligibly.

What is It

POTENTIAL ENERGY AND CONSERVATIVE FORCES

The potential energy concept can be used only when dealing with a special class of forces called *conservative forces*. When only conservative forces act within an isolated system, the kinetic energy gained (or lost) by the system as its members change their relative positions is balanced by an equal loss (or gain) in potential energy. This balancing of the two forms of energy is known as the *principle of conservation of mechanical energy.*

Work is done by a force, and some forces, such as weight, have special characteristics. A *conservative force* is one, like the gravitational force, for which work done by or against it depends only on the starting and ending points of a motion and not on the path taken. We can define a *potential energy* (PE) for any conservative force, just as we did for the gravitational force. For example, when you wind up a toy, an egg timer, or an old-fashioned watch, you do work against its spring and store energy in it. (We treat these springs as ideal, in that we assume there is no friction and no production of thermal energy.) This stored energy is recoverable as work, and it is useful to think of it as potential energy contained in the spring. Indeed, the reason that the spring has this characteristic is that its force is *conservative*. That is, a conservative force results in stored or potential energy. Gravitational potential energy is one example, as is the energy stored in a spring. We will also see how conservative forces are related to the conservation of energy.

An object that possesses kinetic energy can do work on another object—for example, a moving hammer driving a nail into a wall. Now we shall introduce another form of energy. This energy, called **potential energy *U***, is the energy associated with a system of objects.

Gravitational Potential Energy

As an object falls toward the Earth, the Earth exerts a gravitational force ***mg*** on the object, with the direction of the force being the same as the direction of the object's motion. The gravitational force does work on the object and thereby increases the object's kinetic energy.

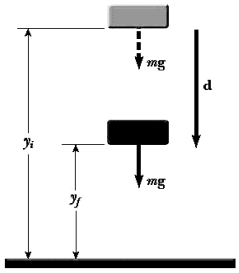
The product of the magnitude of the gravitational force mg acting on an object and the height ***y*** of the object is so important in physics that we give it a name: **the gravitational potential energy**. The symbol for gravitational potential energy is ***Ug*** , and so the defining equation for gravitational potential energy is $U_g \equiv mgy$

Gravitational potential energy is a form of potential energy of the object–Earth system. This potential energy is transformed into kinetic energy of the system by the gravitational force. Thus, the kinetic energy of the system is represented by that of the object falling toward the Earth. Also

note that equation is valid only for objects near the surface of the Earth, where **g** is approximately constant.

Let us now directly relate the work done on an object by the gravitational force to the gravitational potential energy of the object–Earth system. To do this, let us consider a brick of mass **m** at an initial height y_i above the ground, as shown in figure on right. If we neglect air resistance, then the only force that does work on the brick as it falls is the gravitational force exerted on the brick mg. The work **Wg** done by the gravitational force as the brick undergoes a downward displacement ***d*** is

$$W_g = (mg) \cdot d = (-mg\, j) \cdot (y_f - y_i)\, j = mgy_i - mgy_f$$



The work done on the brick by the gravitational force as the brick falls from a height y_i to a height y_f is equal to $mgy_i - mgy_f$.

We just learned that the quantity *mgy* is the gravitational potential energy of the system ***Ug*** , and so we have $W_g = U_i - U_f = -(U_f - U_i) = -\Delta U_g$

From this result, we see that the work done on any object by the gravitational force is equal to the negative of the change in the system's gravitational potential energy.

The unit of gravitational potential energy is the same as that of work—the joule. Potential energy, like work and kinetic energy, is a scalar quantity.

Elastic Potential Energy

Consider a system consisting of a block plus a spring, as shown in figure on right. If the spring is either stretched or compressed a small distance from its unstretched (equilibrium) configuration, it exerts on the block a force of magnitude $F_s = -kx$ where x is the displacement of the block from its unstretched (x=0) position and k is a positive constant called the force constant of the spring. In other words, the force required to stretch or compress a spring is proportional to the amount of stretch or compression x. This force law for springs, known as Hooke's law, is valid only in the limiting case of small displacements. The value of k is a measure of the stiffness of the spring. Stiff springs have large k values, and soft springs have small k values.

Solving for the equation of work done by a spring, we substitute the equation for the force that the spring exerts on the block to $F_s = -kx$ to work done by a variable force equation $W = \int_{X_i}^{X_f} F_x dx$, this will give us the equation $W = \frac{1}{2}kx_i^{\,2} - \frac{1}{2}kx_f^{\,2}$.

In this situation, the initial and final x coordinates of the block are measured from its equilibrium position, *x=0*. Again, we see that *W* depends only on the initial and final *x* coordinates of the object and is zero for any closed path. The elastic potential energy function associated with the system is defined by $U_g \equiv \frac{1}{2}kx^2$

What's More

Sample Problem on Gravitational and Elastic Potential Energy

A 70.0 kg stuntman is attached to a bungee cord with an unstretched length of 15.0 m. He jumps off a bridge spanning a river from a height of 50.0 m. When he finally stops, the cord has a stretched length of 44.0 m. Treat the stuntman as a point mass, and disregard the weight of the bungee cord. Assuming the spring constant of the bungee cord is 71.8 N/m, what is the total potential energy relative to the water when the man stops falling?

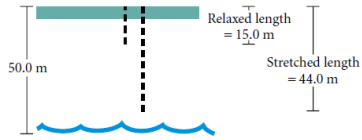
Given: m = 70.0 kg k = 71.8 N/m g = 9.81 m/s2

h = 50.0 m − 44.0 m = 6.0 m x = 44.0 m − 15.0 m = 29.0 m

PE = 0 J at river level

Unknown: PE_{tot} = ?

Diagram:



Choose an equation or situation:

The zero level for gravitational potential energy is chosen to be at the surface of the water. The total potential energy is the sum of the gravitational and elastic potential energy.

$PE_{tot} = PE_g + PE_{elastic}$

$PE_g = mgh$

$PE_{elastic} = \frac{1}{2}kx^2$

Substitute the values into the equations and solve:

PE_g = (70.0 kg)(9.81 m/s2)(6.0 m) = 4.1 × 10^3 J

$PE_{elastic} = \frac{1}{2}$ (71.8 N/m)(29.0 m)^2 = 3.02 × 10^4 J

$PE_{tot} = 4.1 \times 10^3 \text{ J} + 3.02 \times 10^4 \text{ J}$

$\boxed{PE_{tot} = 3.43 \times 10^4 \text{ J}}$

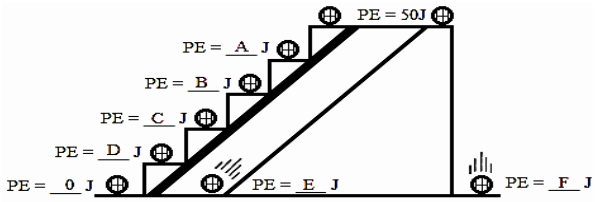
What I Have Learned

Compare and contrast elastic potential energy to gravitational potential energy.

_____

What I Can Do

Use the principle of Gravitational Potential Energy to determine the blanks in the diagram on the right. Knowing that the potential energy at the top of the tall platform is 50 J, what is the potential energy at the other positions shown on the stair steps and the incline?



Assessment

Solve the following problems.

1. A cart is loaded with a brick and pulled at constant speed along an inclined plane to the height of a seat-top. If the mass of the loaded cart is 3.0 kg and the height of the seat top is 0.45 meters, then what is the potential energy of the loaded cart at the height of the seat-top?
2. If a force of 14.7 N is used to drag the loaded cart (from previous question) along the incline for a distance of 0.90 meters, then how much work is done on the loaded cart?

Lesson 3 Conservative Forces

Instead of thinking of interactions as performing work, it is instead possible to think of their effect as transferring kinetic energy to another object or transforming kinetic energy to another form. If we take this view, it is possible to see energy as nature's way of ensuring that "you can't get something for nothing".

In this module we will consider those interactions capable to transform the kinetic energy to another type of energy that can be recovered later on. These types of forces are called conservative forces and the associated energy is called potential energy.

What's In

If frictional forces and air resistance were acting upon the falling ball dropped form top of a building, would the kinetic energy of the ball just prior to striking the ground be more, less, or equal to the kinetic energy of an object in free fall?

What is It

CONSERVATIVE FORCES

A force is conservative if the work it does on a particle moving between two points is independent of the path the particle takes between the two points. Furthermore, a force is conservative if the work it does on a particle is zero when the particle moves through an arbitrary closed path and returns to its initial position.

Conservative forces have two important properties:

1. A force is conservative if the work it does on a particle moving between any two points is independent of the path taken by the particle.
2. The work done by a conservative force on a particle moving through any closed path is zero. (A closed path is one in which the beginning and end points are identical.)

The gravitational force is one example of a conservative force, and the force that a spring exerts on any object attached to the spring is another. As we learned in the preceding section, the work done by the gravitational force on an object moving between any two points near the Earth's surface is $Wg = mgy_i - mgy_f$. From this equation we see that Wg depends only on the initial and final y coordinates of the object and hence is independent of the path. Furthermore, Wg is zero when the object moves over any closed path (where $y_i = y_f$)

For the case of the object–spring system, the work Ws done by the spring force is given by $Ws = \frac{1}{2}kx_i^2 - \frac{1}{2}kx_f^2$. Again, we see that the spring force is conservative because Ws depends only on the initial and final x coordinates of the object and is zero for any closed path.

We can associate a potential energy with any conservative force and can do this only for conservative forces. In the previous section, the potential energy associated with the gravitational force was defined as $U_g \equiv mgy$**.** In general, the work Wc done on an object by a conservative force is equal to the initial value of the potential energy associated with the object minus the final value:

$$W_c = U_i - U_f = -\Delta U$$

This equation should look familiar to you. It is the general form of the equation for work done by the gravitational force and that for the work done by the spring force.

Let us now consider what form the work-energy theorem takes when only conservative forces are involved. This will lead us to the conservation of energy principle. The work-energy theorem states that the net work done by all forces acting on a system equals its change in kinetic energy. In equation form, this is $W_{net} = \frac{1}{2}mv^2 - \frac{1}{2}mv_o^2 = \Delta KE$

If only conservative forces act, then $W_{net} = W_c$

where Wc is the total work done by all conservative forces. Thus, $W_c = \Delta KE$

Now, if the conservative force, such as the gravitational force or a spring force, does work, the system loses potential energy. That is, $W_c = -\Delta PE$**.** Therefore, $-\Delta PE = \Delta KE$ or $\Delta PE + \Delta KE = 0$

What's More

Conservation of Mechanical Energy

The mechanical energy is the sum of kinetic energy and all forms of potential energy associated with an object or group of objects.

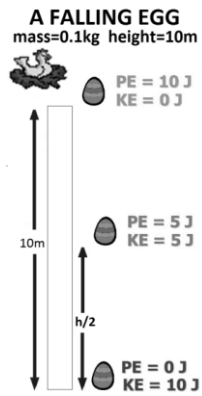
$$ME = KE + PE_{tot}$$

In the absence of friction, the total mechanical energy remains the same. This principle is called *conservation of mechanical energy*. Although the amount of mechanical energy is constant, mechanical energy itself can change form.

For instance, consider the forms of energy for the falling egg, as shown. As the egg falls, the potential energy is continuously converted into kinetic energy. If the egg were thrown up in the air, kinetic energy would be converted into gravitational potential energy. In either case,

mechanical energy is conserved. The conservation of mechanical energy in the absence of air friction can be written symbolically as follows: $ME_i = ME_f$

$$initial\ mechanical\ energy = final\ mechanical\ energy$$



The mathematical expression for the conservation of mechanical energy depends on the forms of potential energy in a given problem. For instance, if the only force acting on an object is the force of gravity, as in the egg example, the conservation law can be written as follows:

$$\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2 + mgh_f$$

Sample Problem: Starting from rest, a child zooms down a frictionless slide from an initial height of 3.00 what is her speed at the bottom of the slide? Assume she has a mass of 25.0 kg.

Given:

$h = hi = 3.00$ m $\quad m = 25.0$ kg $\quad v_i = 0.0$ m/s $\quad h_f = 0$ m

Unknown: v_f =?

Solution:

The slide is frictionless, so mechanical energy is conserved.

Kinetic energy and gravitational potential energy are the only forms of energy present.

$$KE = \frac{1}{2}mv^2 \qquad PE = mgh$$

The zero-level chosen for gravitational potential energy is the bottom of the slide. Because the child ends at the zero level, the final gravitational potential energy is zero.

$$PE_{g,f} = 0$$

The initial gravitational potential energy at the top of the slide is $PE_{g,i} = mgh_i = mgh$

Because the child starts at rest, the initial kinetic energy at the top is zero.

$$KE_i = 0$$

Therefore, the final kinetic energy is as follows:

$$KE_f = \frac{1}{2}mv_f^2$$

Substitute values into the equations:

$$PE_{g,i} = (25kg)(9.81m/s^2)(3.00m) = 736\,J$$

$$KE_f = \frac{1}{2}(25.0kg)v_f^2$$

Now use the calculated quantities to evaluate the final velocity.

$$ME_i = ME_f$$

$$PE_i + KE_i = PE_f + KE_f$$

$$736\,J + 0\,J = 0\,J + (0.500)(25.0kg)\,v_f^2$$

What I Have Learned

$$v_f = 7.67\,m/s$$

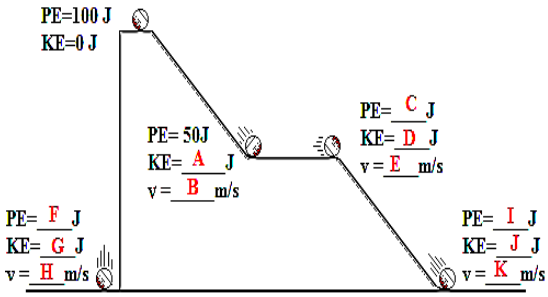
What is the relationship of potential energy and conservative force?

_____

_____

What I Can Do

Consider the falling and rolling motion of the ball in the following two resistance-free situations. In one situation, the ball falls off the top of the platform to the floor. In the other situation, the ball rolls from the top of the platform along the staircase-like pathway to the floor. For each situation, indicate what types of forces are doing work upon the ball. Is the energy of the ball conserved? ?hy? Finally, fill in the blanks for the 2-kg ball.



Assessment

Check your understanding, answer the following questions.

1. The force exerted by a diving board is conservative, provided the internal friction is negligible. Describe changes in the potential energy of a diving board just before the swimmer steps on the board until just after his feet leave it.
2. What is the relationship of potential energy to conservative force?

Potential Energy Diagram and Center of Mass
Quarter 1 – Week 7

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topics on potential energy diagram and center of mass. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using. The module has two lessons, namely:

- Lesson 1 – Potential Energy Diagram
- Lesson 2 – Center of Mass

After going through this module, you are expected to:

1. use potential energy diagrams to infer force; stable, unstable, and neutral equilibria;
2. solve problems involving work, energy, and power; and
3. differentiate center of mass and geometric center.

What I Know

Directions: Read each statement carefully. On the space provided before each number, write FACT if the statement is correct. Otherwise, write BLUFF.

_____ 1. Potential energy is the energy stored in an object due to its position.
_____ 2. Both gravitational and elastic potential energy are dependent upon the mass of an object.
_____ 3. If the mass of an elevated object is doubled, then its gravitational potential energy will be doubled as well.
_____ 4. The unit of measurement for potential energy is the Joule.
_____ 5. The higher that an object is, the more potential energy which it will have.

Lesson 1 Potential Energy Diagram

Often, you can get a good deal of useful information about the dynamical behavior of a mechanical system just by interpreting a graph of its potential energy as a function of position, which is called a ***potential energy diagram***.

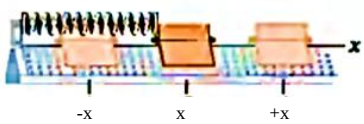
What's In

What is the equilibrium position? *And how it relates to the stable, unstable and neutral equilibrium?*

What is It

Having a block-spring system (Fig. 1) at rest or a person sitting idle on a swing is considered at the ***equilibrium position***. Both systems will remain at rest (x=0) in the absence of an external force. If the spring is stretched (+x), both the x and slope are positive, while the force is negative, since the force acts as a restoring force on the spring by moving towards the equilibrium position. If the spring is compressed, the x and slope will be negative, while the force is positive.

Fig. 1 **Block-spring system**
Source Young, H. D., Freedman, R. A., Ford, A. L., & Sears, F. W. (2004). *Sears and Zemansky's university physics: With modern physics*. San Francisco: Pearson Addison



Considering a spring system by using the elastic potential energy, the force exerted on the spring will yield the equation.

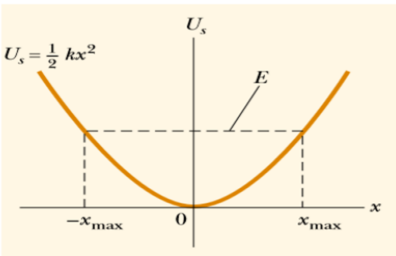
$$F_S = \frac{-dU_S}{dx} = -kx$$

where:

F_s = Force on the spring U_s = Elastic Potential energy
x = displacement k = Spring constant

Energy, force and equilibrium can be described qualitatively by graphing the potential energy versus x curve.

Fig. 2 **Stable Equilibrium**
Source Serway, R. A., Jewett, J. W., & Serway, R. A. (2004). *Physics for scientists and engineers*. Belmont, CA: Thomson-Brooks/Cole.



In Fig. 2, we see that if the block is given an initial displacement +*xmax*, its potential energy stored in the spring intersects with the total energy in the system (E). If we were to release the spring from +*xmax*, as the block starts to move, the system acquires kinetic energy equal to the amount of lost potential energy, because the total energy of the system must remain constant, thus kinetic energy reaches its peak where the potential energy is at its lowest. The conservative force here acts to push the spring toward a lower potential energy.

When oscillating the block reaches the highest points +*xmax* and -*xmax*, it changes direction and moves toward the equilibrium position, since these are called *turning points*, the block must move back in order to conserve energy. As shown in fig.2, The horizontal line represents the *Total Mechanical Energy(E)* of the system. If the force on the spring causes it to oscillate, eventually resulting in the block returning to its equilibrium position is called ***stable equilibrium***.

Now consider the equilibrium position of a pen by standing on its tip, can it maintain this equilibrium position without your helping hand? The moment your hand leaves or its slightly displaced it would fall on either side, and this is called ***unstable equilibrium***.

Now consider a particle moving along the x axis under the influence of a conservative force F_x, where the U versus x curve is as shown in fig. 3. Once again, $F_x = 0$ at the equilibrium position at x = 0. However, this is a position of unstable equilibrium for the following reason: Suppose that

the particle is displaced to the right. Because the slope is negative, the force is positive and the particle accelerates away from the equilibrium position. The force will not return to the equilibrium position instead it will place the object at a lower potential energy. In general, any maximum position in a U versus x curve is in unstable equilibrium.

Finally, a situation may arise where U is constant over some region. This is called a configuration of ***neutral equilibrium***. Small displacements from a position in this region produce neither restoring nor disrupting forces. A ball lying on a flat horizontal surface is an example of an object in neutral equilibrium.

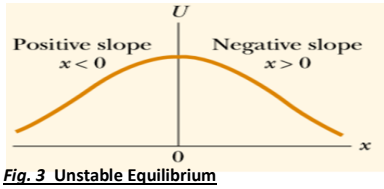


Fig. 3 **Unstable Equilibrium**
Source Serway, R. A., Jewett, J. W., & Serway, R. A. (2004). *Physics for scientists and engineers*. Belmont, CA: Thomson-Brooks/Cole.

What's More

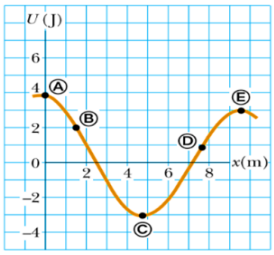
Activity: Swing it!

Materials: String, bottle cap

Part A.

1. Make a simple pendulum with string and bottle cap.
2. Now stretch the string and hold the bottle cap at 60° from the -y axis.
3. As it swings to a stop, draw (in a separate paper) and label at which point of it travels:
   a. the highest and lowest points for KE and PE
   b. equilibrium position
   c. turning points
4. With the given materials, think of a way to produce unstable equilibrium.
5. With the given materials, think of a way to produce neutral equilibrium.

Fig. 4 **Potential Energy versus x-curve #1**
Source Serway, R. A., Jewett, J. W., & Serway, R. A. (2004). *Physics for scientists and engineers*. Belmont, CA: Thomson-Brooks/Cole.



Guided Questions:

Answer the following questions explicitly. Answer on a separate sheet

1. What type of equilibrium is observed in the pendulum swinging to a stop? Why?
2. State why your 2nd set-up produces unstable equilibrium.
3. State why your 3rd set-up produces neutral equilibrium.

Part B.

A particle moves along a line where the potential energy of its system depends on its position x as graphed in the fig. 4.

a) Identify for each labelled position of this particle, whether each position is of stable, unstable, or neutral equilibrium.
b) At which of the labeled x-coordinates is the force on the particle zero?

What I Have Learned

Differentiate stable equilibrium, unstable equilibrium and neutral equilibrium.

What I Can Do

Answer the following question on a separate sheet of paper.

Trivia: The Three Gorges Dam is the largest hydroelectric dam in the world which spans over the Yangtze river in China. The Yangtze river is the longest river in Asia but ranks 3rd in the world.

1. Trace the different forms of energy present in the generation of electricity on hydroelectric dam? _____

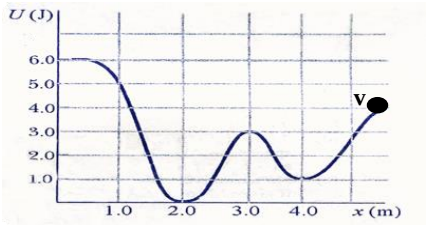
2. Can the generated electrical energy exceed the stored potential energy? _____

Assessment

Answer the following questions explicitly.

1. A right circular cone can be balanced on a horizontal surface in three different ways. Sketch these three equilibrium configurations, and identify them as positions of stable, unstable, or neutral equilibrium on the box provided.

Fig. 5 **Potential energy vs x-curve #2**
Source. https://www.homeworklib.com



2. For the potential energy curve shown in Figure 5, Assume at the end of the graph beyond point 4.0 m the graph is constant at 4.0 J.
a) Indicate points of stable, unstable, and neutral equilibrium on the graph.
b) If the system has a total energy of 4 J, what is the maximum kinetic energy?
c) Sketch the curve for F_x versus x from x = 0 to x = 4.0 m on the graph.

Lesson 2 Center of Mass

What is It

For a given body, the **center of mass** is the average position of all the mass that makes up the object. For example, Fig. 1 show the center of mass for *regularly shaped* objects at its geometric center. Things will seem different for *irregularly shaped* objects (Fig. 2), the center of mass would be located outside the object.

When throwing an irregular shaped object (Fig. 3), the center of mass will always be at the *trajectory* while the rest of the body rotates.

Fig. 1 **Center of mass for regular shaped objects**
Source http://www.citycollegiate.com/centre_of_mass.htm

Fig. 2 **Center of mass for irregular shaped objects**
Source www.canva.com

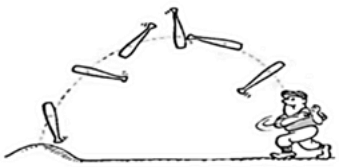


Fig. 3 **Center of mass of a thrown irregular shaped object**
Source Hewitt. P. G. (2006). *Conceptual physics*. 10th ed.

What's More

Materials: 3 short bond papers, scissors, pen
Procedures:

1. Make 3 different shaped objects (e.g., stars, box or a map)
2. Pick a random point at the edge of your cut out and suspend it.
3. From that point, draw a straight line downward.
4. Repeat steps 2 & 3 twice.

Guide Question:
What does the intersection of lines represent?
_____

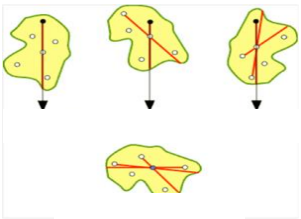


Fig. 4 Source :https://www.lsu.edu/physics/resources/lecture-demonstrations/mechanics/static-rigid-bodies.php

What I Have Learned

What is the difference between center of mass from geometric center? How can we determine the geometric center of an irregular shaped object?

_____

What I Can Do

Answer the following question.

1. Can an object have more than one center of mass? Why?
   _____
2. Where is the center of mass of Earth's atmosphere?
   _____

Assessment

On a separate sheet of paper, answer the following questions explicitly.

1. Draw/print the Philippine archipelago. (Rough sketch would suffice)
2. Locate the center of mass of the Philippine archipelago.

Impulse, Momentum and Collision Problems
Quarter 1 – Week 8

Lesson 1-3	Collisions, Impulse and Momentum

What is It

The interaction between your slippers and the can when you play *tumbang preso*, a ball and your hand when you play volleyball, and as the cue ball hits the other billiard balls are some examples of collisions. This will be our focus for this lesson. But first, let us define collision. A **collision** is an interaction between two or more objects in a short period of time. As collision happens, change between the initial and final velocities of each object is evident. Moreover, the

time of contact between the two objects depends what made up the material is made. A large force exerted during the time of contact is called an ***impulsive***, *F(t)*. If we will be trying to analyze collision using Newton's Second Law,

$$ma_x = m\frac{dv_x}{dt} = F_x(t)\ (eq.1)$$

Rearranging the equation:

$$mdv_x = F_x(t)dt\ (eq.2)$$

We can integrate equation 2 since the time interval between t_i and t_f is nonzero which results to

$$m\int_{v_i}^{v_f} dv_x = mv_{xi} - mv_{xf} = \int_{t_i}^{t_f} F_x(t)dt\ (eq.3)$$

What's More

Momentum refers to the product of the particle's mass and velocity. It is also a vector quantity with an SI unit 1 kg m/s. It is represented in a mathematical equation p=mv (eq. 4) where p = momentum, m = mass, and v = velocity. Analyzing the equation, an object with a small mass but with a large velocity or a large mass with a small velocity can have a large momentum.

Since momentum is a vector quantity, it has x- and y- components as well and it is expressed on the equation below:

$$p_x = mv_x\ (eq.5) p_y = mv_y\ (eq.6)$$

Momentum in terms of Newton's Second Law of Motion is

$$F = ma = m\frac{dv}{dt} = \frac{d\ (mv)}{dt} = \frac{dp}{dt}\ (eq.7)$$

This equation implies that *force is the rate of change of momentum.*

Analyzing equation 3, we can say that mv_{fx} *and* mv_{fi} are just the initial and final momentum of a body, respectively. Therefore, in terms of momentum, we can express equation 3 as

$$\Delta p_x = p_{xf} - p_{xi} = \int_{t_i}^{t_f} F_x(t)\ dt\ (eq.8)$$

Impulse

Equation 8 describes particles' change in momentum and its relation to the time integral of the force. With this, we can define impulse (J) as

$$J_x = \int_{t_i}^{t_f} F_x(t)\ dt\ (eq.9)$$

Analyzing the equation, the SI unit of impulse is 1 1 N s which is also equal to 1 kg m/s, the SI unit of momentum.

Impulse – Momentum Theorem

Integrating equation 3, we can find out that Newton's second law of motion can be written in terms of impulse and momentum as

$$\Delta p = J_x\ (impulse - momentum\ theorem)(eq.10)$$

This equation tells us that **an impulse experienced by a particle is equal to the change in its momentum.**

Law of Conservation of Momentum states that the total momentum P of an isolated system is a constant. Interactions within the system does not change the system's total momentum.

This law can be expressed in a mathematical equation

$$p_f = p_i\ (momentum\ before\ and\ after\ the\ collision)\ (eq.11)$$

Types of Collisions

In every collision, *momentum is being conserved*. However, some of these collisions do not conserve kinetic energy.

Elastic Collisions – This type of collision conserves kinetic energy. In other words, the total kinetic energy before and after the collision does not change or being transformed into other types of energy. This is explained in the equations that follow:

$$p_f = p_i \text{ (momentum before and after the collision)} m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$where: m_1 = mass\ of\ object\ 1;\ m_2 = mass\ of\ object\ 2;\ v_{1i} = velocity\ of\ object\ 1\ before\ the\ collision;$$

$$v_{2i} = velocity\ of\ the\ object\ 2\ before\ the\ collision;$$

$$v_{1f} = velocity\ of object\ 1\ after\ the\ collision;$$

$$and\ v_{2f} = velocity\ of\ object\ 2\ after\ the\ collision$$

Applying the conservation of Kinetic energy,

$$\frac{1}{2}m_1 {v_{1i}}^2 + \frac{1}{2}m_2 {v_{2i}}^2 = \frac{1}{2}m_1 {v_{1f}}^2 + \frac{1}{2}m_2 {v_{2f}}^2$$

Inelastic Collisions – Objects that stick together after the collision and move in the same direction are experiencing inelastic collision. The total momentum is being conserved but the total kinetic energy is not. In this type of collision, the kinetic energy had been converted into other kinds of energy such as heat, or sound. The following equations further explain inelastic collisions.

$$p_f = p_i \text{ (momentum before and after the collision)} m_1 v_{1i} + m_2 v_{2i} = v_f (m_1 + m_2)$$

$$where: m_1 = mass\ of\ object\ 1;\ m_2 = mass\ of\ object\ 2;\ v_{1i} = velocity\ of\ object\ 1\ before\ the\ collision;$$

$$v_{2i} = velocity\ of\ the\ object\ 2\ before\ the\ collision;\ and$$

$$v_f = velocity\ of object\ 1\ and\ object\ 2\ after\ the\ collision$$

The kinetic energy before and after the collision,

$$\frac{1}{2}m_1 {v_{1i}}^2 + \frac{1}{2}m_2 {v_{2i}}^2 = \frac{1}{2}(m_1 + m_2)\ {v_{1f}}^2 + Q$$

'Q' represents the change in energy as a result of production of heat or sound.

The Coefficient of Restitution

The coefficient of restitution (e) is the ratio of the relative velocity of the objects colliding before and after the collision. This is also the consequence of Newton's Third Law of Motion. If the value of e is 1, the collision is said to be elastic. Moreover, if the value of e is 0, it is an inelastic collision.

Coefficient of restitution (e) uses the equations that follow:

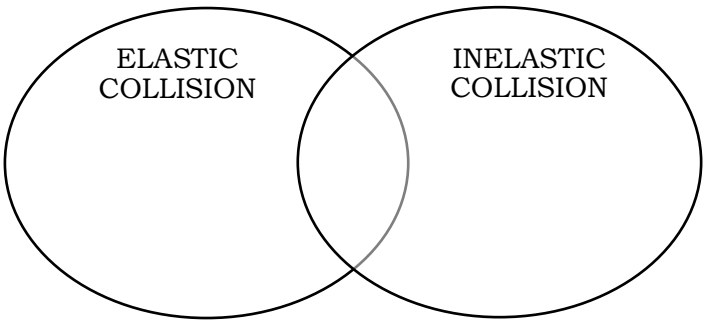
$$e = \left|\frac{v_B - v_A}{u_B - u_A}\right|$$

where: v_B – velocity of object B after the collision
v_A – velocity of object A after the collision
u_B – velocity of object B before the collision
u_A – velocity of object A before the collision

Activity: The Impact!

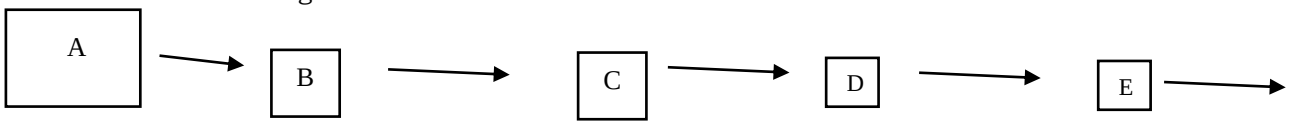
A. Elastic and Inelastic Collision

Using the Venn diagram below, compare and contrast elastic and inelastic collisions.



B. Momentum and Impulse

1. Rank from largest to smallest the momenta from block A to E.



A	B	C	D	E
m= 200 g	m= 20 g	m= 20 g	m= 10 g	m= 10 g
v = 0.1 m/s	v = 2 m/s	v = 1 m/s	v= 2 m/s	v= 1m/s

Order: _____

2. A carnival game requires you to knock over a wood post by throwing a ball on it. You are offered with a very bouncy ball and a very sticky clay ball of equal mass. Assume that you can throw them with equal speed and equal accuracy. You only get one to choose. Which ball will you choose? Why?
   _____
3. Two particles collide, one of which was initially moving and the other initially at rest.
   a. Is it possible for *both* particles to be at rest after the collision? Justify your answer by providing examples.
   _____
   b. Is it possible for *one* particle to be at rest after the collision? Justify your answer by providing examples. _____
4. Solve the problem below:
   A 480-kg car moving at 14.4 m/s hits from behind a 570-kg car moving at 13.3 m/s in the same direction. If the new speed of the heavier car is 14.0 m/s, what is the speed of the lighter car after the collision, assuming that any unbalanced forces on the system are negligibly small?

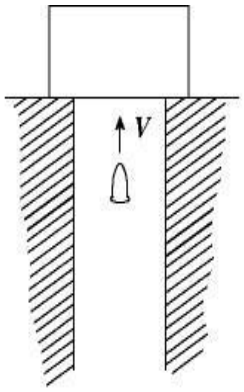
What I Have Learned

How do elastic collisions differ from inelastic collisions? What does restitution of coefficient of collisions mean? What does impulse – momentum theorem imply?

_____

_____

What I Can Do



Analyze the situation and complete the statement below
A 15-g bullet is shot vertically into a 2-kg block. The block lifts upward 8.0 mm (see the figure). The bullet penetrates the block and comes to rest in it in a time interval of 0.0010 s. Assume the force on the bullet is constant during penetration and that air resistance is negligible. The initial kinetic energy of the bullet is closest to _____.

Assessment

***Directions**: Write the letter of your choice on the space provided before each number. (USE CAPITAL LETTER ONLY)*

_____1. Which of the following refers to the interaction between objects on a short period of time?
   A. collision C. momentum
   B. impulse D. forces

_____2. On all types of collision, which quantity is being conserved?
   A. impulse C. momentum
   B. kinetic energy D. potential energy

_____ 3. Analyze the two statements below:

I. Elastic collisions conserve both momentum and kinetic energy.

II. Inelastic collisions conserve kinetic energy but not momentum.

A. Both statements are true.
B. Both statements are false.
C. Only statement I is true.
D. Only statement II is true.

_____ 4. During a collision with a wall, the velocity of a 0.200-kg ball changes from 20.0 m/s toward the wall to 12.0 m/s away from the wall. If the time the ball was in contact with the wall was 60.0 m/s, what was the magnitude of the average force applied to the ball?

A. 13.3 N
B. 40.0 N
C. 26.7 N
D. 107 N

_____ 5. A car of mass 1689 kg collides head-on with a parked truck of mass 2000 kg. Spring mounted bumpers ensure that the collision is essentially elastic. If the velocity of the truck is 17 km/hr (in the same direction as the car's initial velocity) after the collision, what was the initial speed of the car?

A. 29 km/h
B. 38 km/hr
C. 19 km/hr
D. 10 km/hr

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